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Sleep Medicine 2026

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November 2026

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Introduction

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- Normal human sleep
- Sleep and aging
- Polysomnography
- Sleep Apnea
- Restless Leg Syndrome / Periodic Leg Movements
- Narcolepsy
- Insomnia

- ***Thank you to Dr. Waxner for the background slides

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Normal Human Sleep

Normal Human Sleep

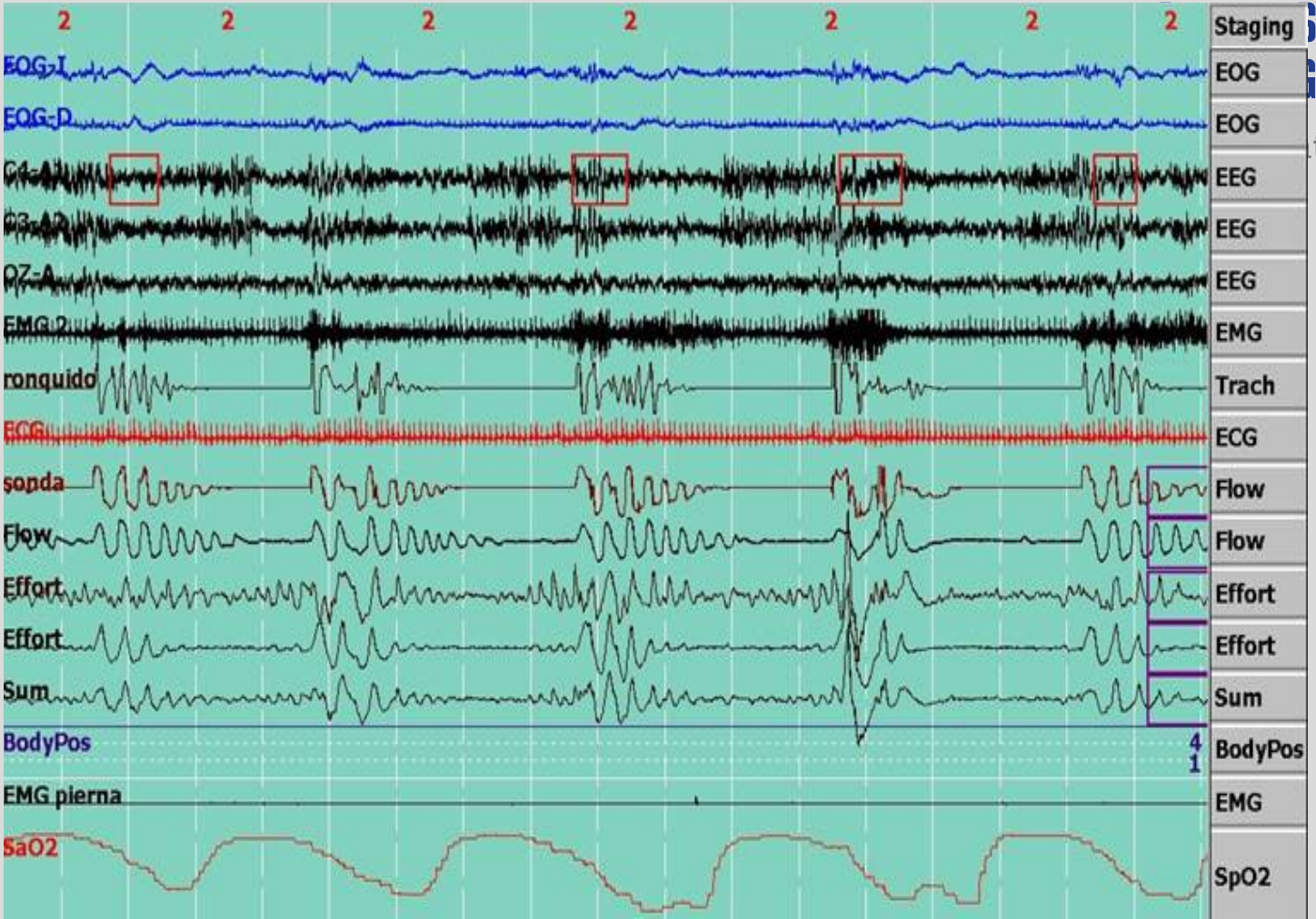
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- Normal human sleep comprises two states-non-rapid eye movement (NREM) and REM sleep-that alternate cyclically across a sleep episode
- **REM sleep:**
  - muscles are atonic
  - dreaming is typical
  - bursts of rapid eye movements

- Sleep begins in NREM and progresses through deeper NREM stages (stages N1, N2, N3/4 (SWS)) before the first episode of REM sleep approximately 70 to 100 minutes later.
- Thereafter, NREM sleep and REM sleep cycle with a period of approximately 90 minutes



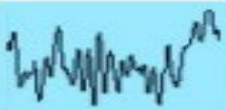
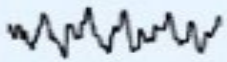
OVERNIGHT POLYSOMNOGRAM



Normal Human Sleep

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- **EEG CHARACTERISTICS OF NREM AND REM**
  - AWAKE – eyes closed - ALPHA Waves 8-12 Hz
  - STAGE 1 - THETA Waves 4-8 Hz
  - STAGE 2 - K COMPLEXES and SPINDLES
  - SWS (Slow Wave Sleep) - DELTA Waves (< 4 Hz) >20% of epoch. Stage 3 and 4 combined now.
  - REM similar to awake state: SAWTOOTH WAVES

Sample	Label	Definition
	Alpha activity	8–13 Hz rhythm, usually most prominent in occipital leads. Thought to be generated by cortex, possibly via dipole located in layers 4 and 5. Used as a marker for relaxed wakefulness and CNS arousals.
	Theta activity	4–8 Hz waves, typically prominent in central and temporal leads. Sawtooth activity (shown in figure) is a unique variant of theta activity (containing waveforms with a notched or *sawtooth-shaped* appearance) frequently seen during REM sleep.
	Vertex sharp waves	Sharply contoured, negative-going bursts that stand out from the background activity and appear most often in central leads placed near the midline.
	Sleep spindle	A phasic burst of 11–16 Hz activity, prominent in central scalp leads; typically last for 0.5–1.5 seconds. Spindles are a scalp representation of thalamocortical discharges; the name derives from their shape (which is spindle-like).
	K complex	Recently redefined in the AASM manual as *an EEG event consisting of a well-delineated negative sharp wave immediately followed by a positive component standing out from the background EEG with total duration ≥ 0.5 seconds, usually maximal in amplitude over the frontal regions.*
	Slow waves	High-amplitude (≥ 75 μvolts) and low-frequency (≤ 2 Hz) variants of delta (1–4 Hz) activity. Slow waves are the defining characteristics of stage N3 sleep.
	REM	Rapid eye movements are conjugate saccades occurring during REM sleep correlated with the dreamer's attempt to look at the dream sensorium. They are sharply peaked with an initial deflection usually < ½ second in duration.
	SEM	Slow eye movements are conjugate, usually rhythmic, rolling eye movements with an initial deflection usually ≥ ½ second in duration.

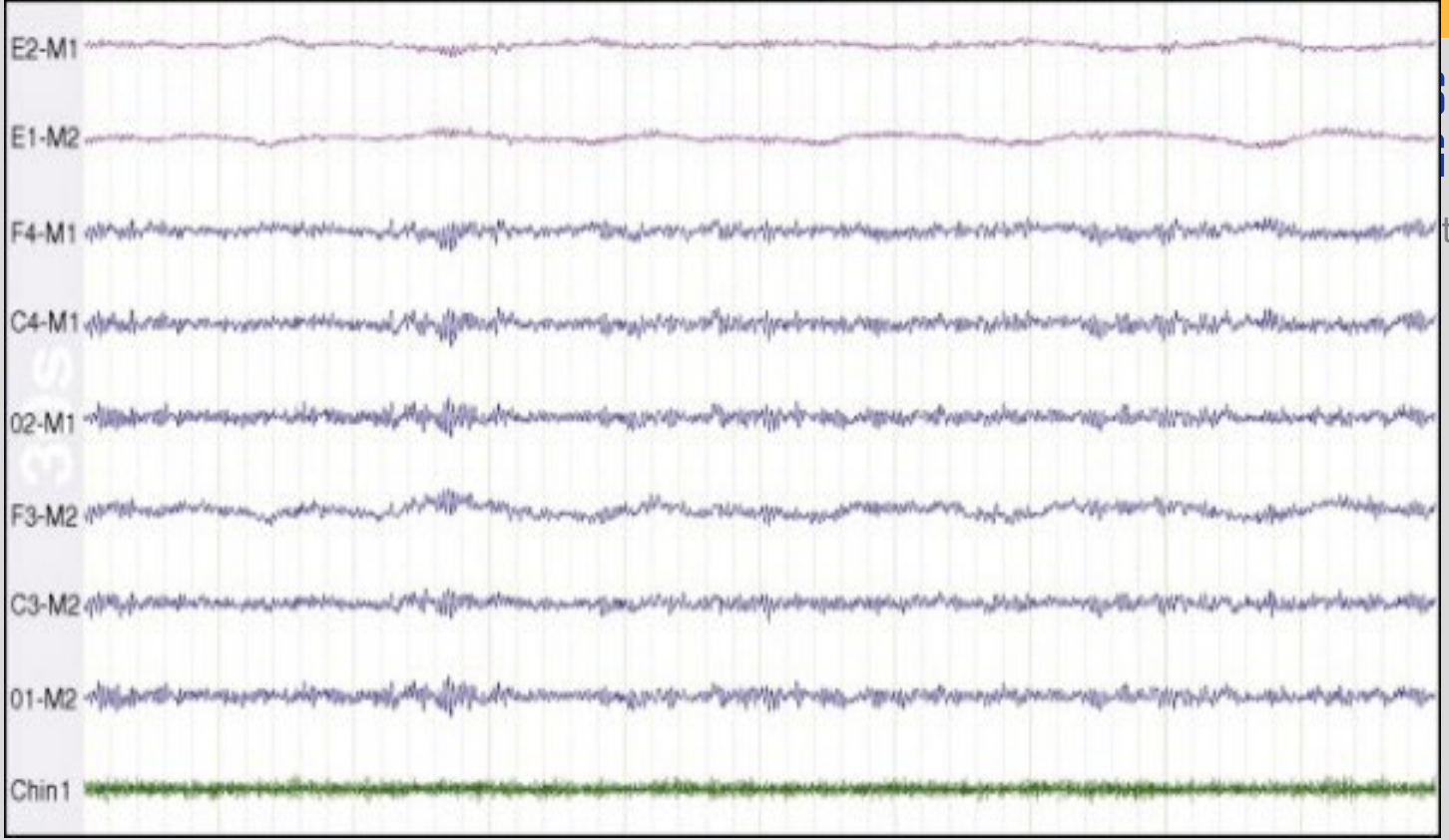


Figure 4-2 Stage W (wakefulness). This fragment is a 30-second epoch. The two top channels represent the channels that monitor eye movements. The next six channels are used to record the EEG. The first three of these (F4-M1, C4-M1, O2-M1) are the ones recommended for scoring sleep. The others are backups in case electrodes malfunction. The bottom channel is the chin EMG. The key finding that indicates this is wakefulness is the presence of alpha rhythm over the occipital region (monitored by the O leads) occupying more than 30 seconds of the epoch. See , , for technical details and definitions of the waves seen in sleep. to use the same montage.

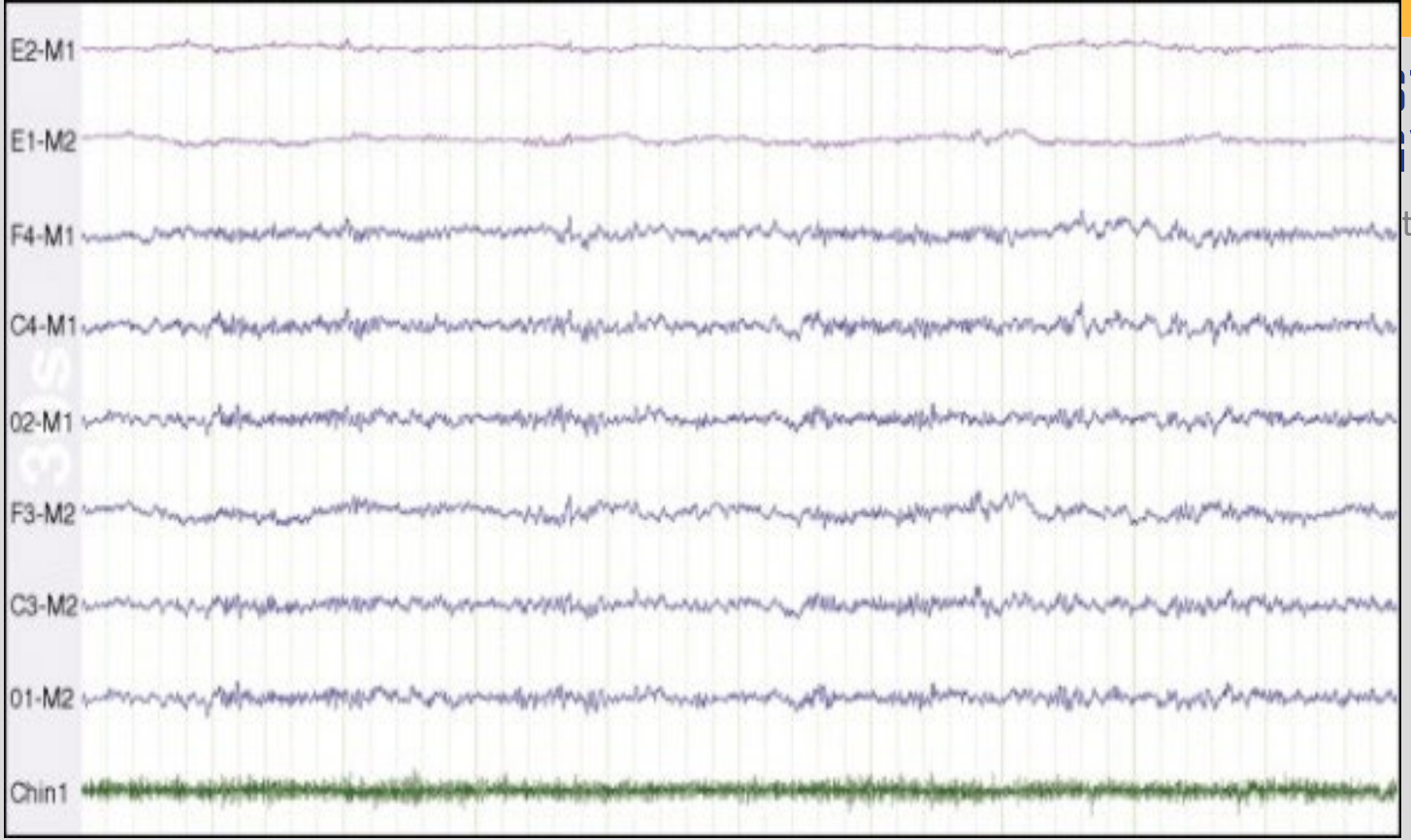


Figure 4-3 Stage N1. This is stage N1 because a mixed frequency, low-amplitude pattern has replaced the alpha rhythm for more than 50% of the epoch

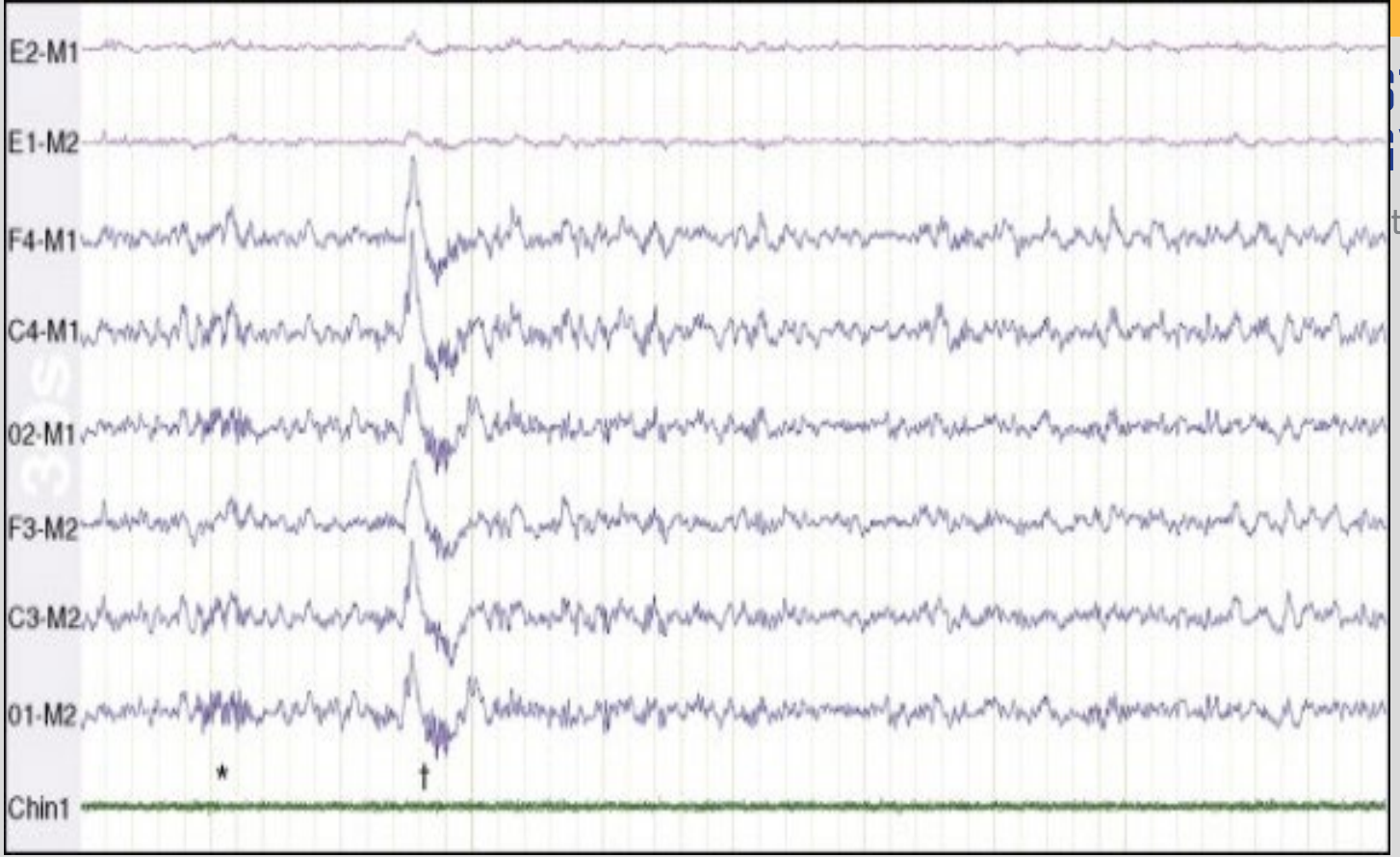


Figure 4-4 Stage N2. A train of sleep spindles (*) toward the beginning of the epoch is followed shortly by a K complex (†).

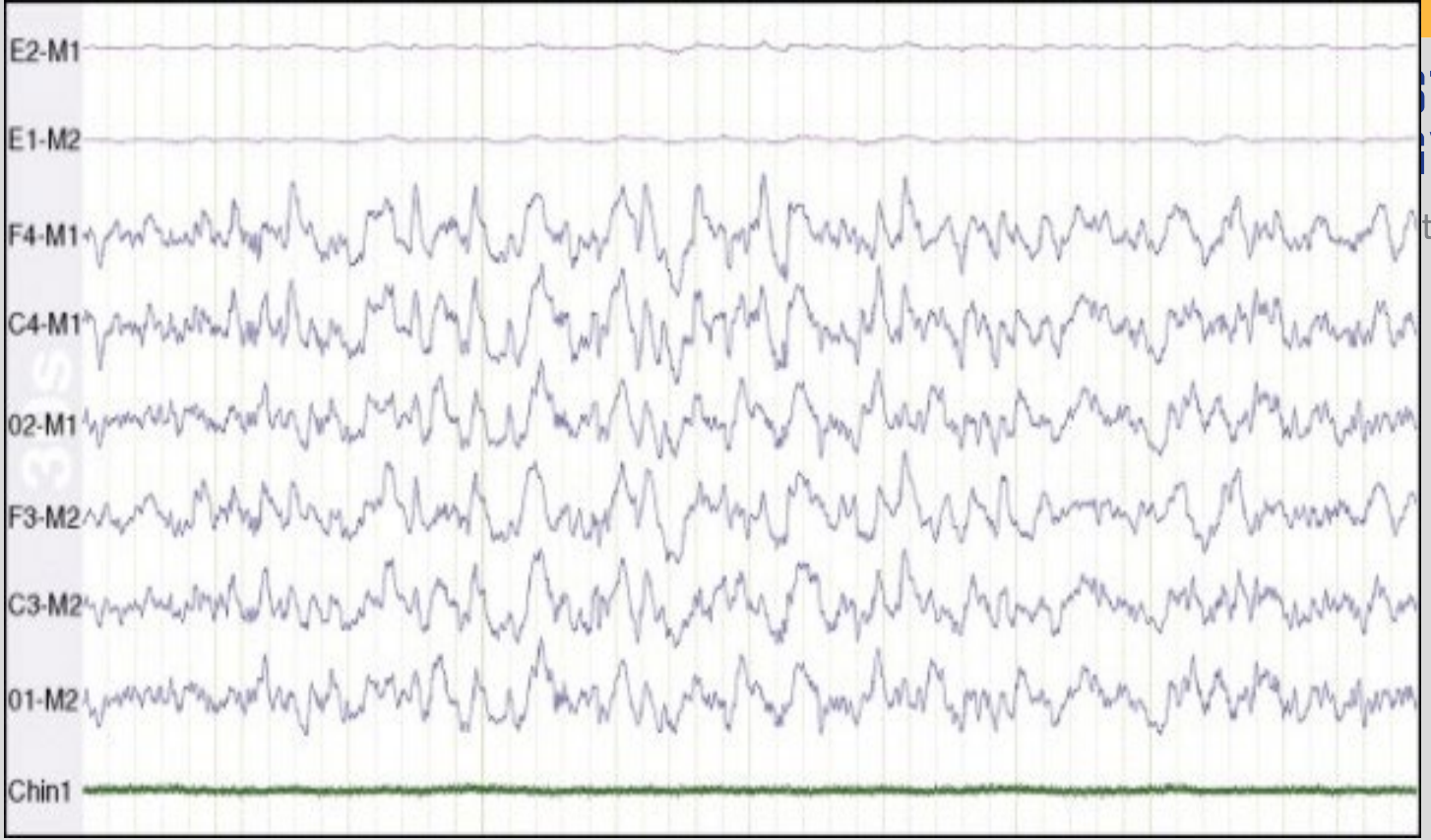


Figure 4-5 Stage N3. This is stage N3 because more than 20% of the epoch consists of slow-wave activity.



Figure 4-6 Stage R. There are rapid eye movements (*) and a mixed-frequency, low-amplitude EEG; muscle tone (chin EMG) is absent (†). Note that some of the activity of the eyes has spilled over into the two frontal (F) leads (#)

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Sleep and Aging

Normal Human Sleep and Aging

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- **Age-related changes are predictable:**
  - Newborns enter REM sleep (called *active sleep*) before NREM (called *quiet sleep*) and have a shorter sleep cycle (approximately 50 minutes)
  - Sleep stages emerge as the brain matures during the first year
  - At birth, active sleep is approximately 50% of total sleep and declines over the first 2 years to approximately 20% to 25%
  - Slow wave sleep (stages 3 and 4) decreases across adolescence by 40% from preteen years and **continues to decline into old age**, particularly in men and less so in women
  - REM sleep as a percentage of total sleep is approximately 20% to 25% across childhood, adolescence, adulthood, and into old age except in dementia.

Normal Human Sleep

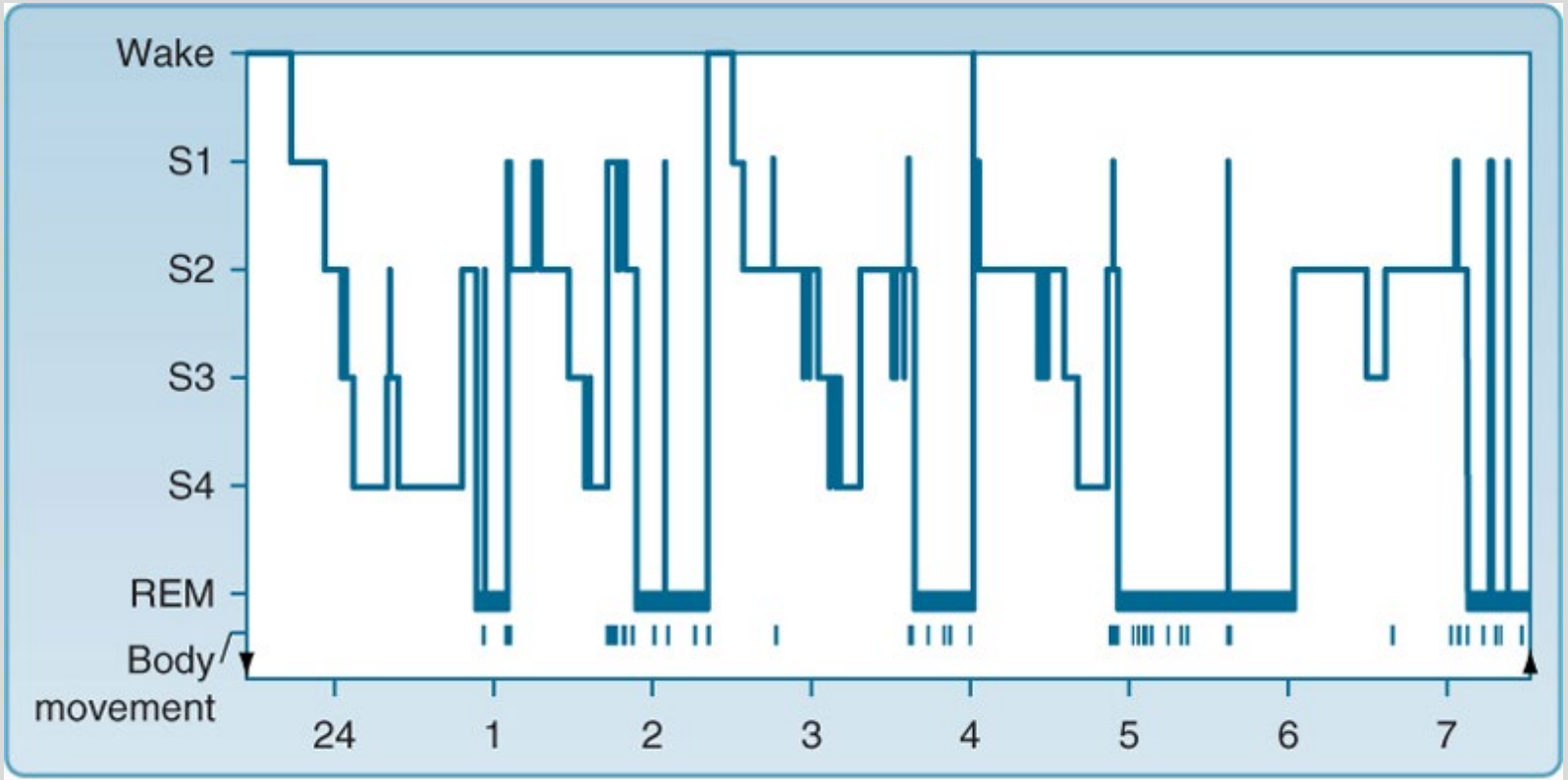
- ***Generalizations about Sleep in the Normal Young Adult***
  - Sleep is entered through NREM sleep
  - NREM sleep and REM sleep alternate with a period near 90 minutes
  - SWS predominates in the first third of the night and is linked to the initiation of sleep
  - REM sleep predominates in the last third of the night and is linked to the circadian rhythm of body temperature

Normal Human Sleep

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- ***Generalizations about Sleep in the Normal Young Adult***
  - Wakefulness in sleep usually accounts for less than 5% of the night
  - Stage 1 sleep generally constitutes approximately 2% to 5% of sleep
  - Stage 2 sleep generally constitutes approximately 45% to 55% of sleep
  - SWS sleep generally constitutes approximately 13% to 23% of sleep
  - NREM sleep, therefore, is usually 75% to 80% of sleep
  - REM sleep is usually 20% to 25% of sleep, occurring in four to six discrete episodes.

Normal Human Sleep



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Normal Human Sleep and Aging

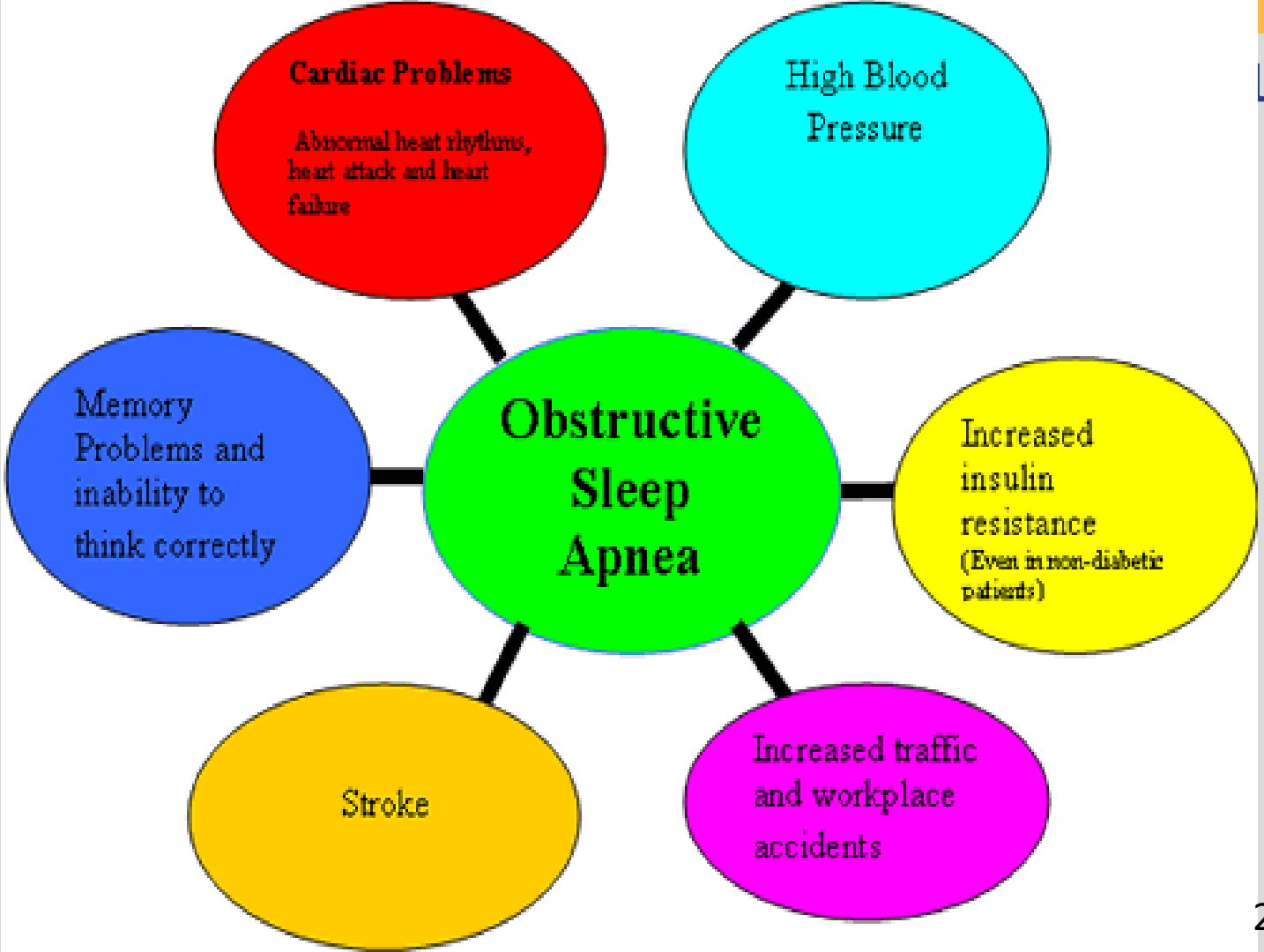
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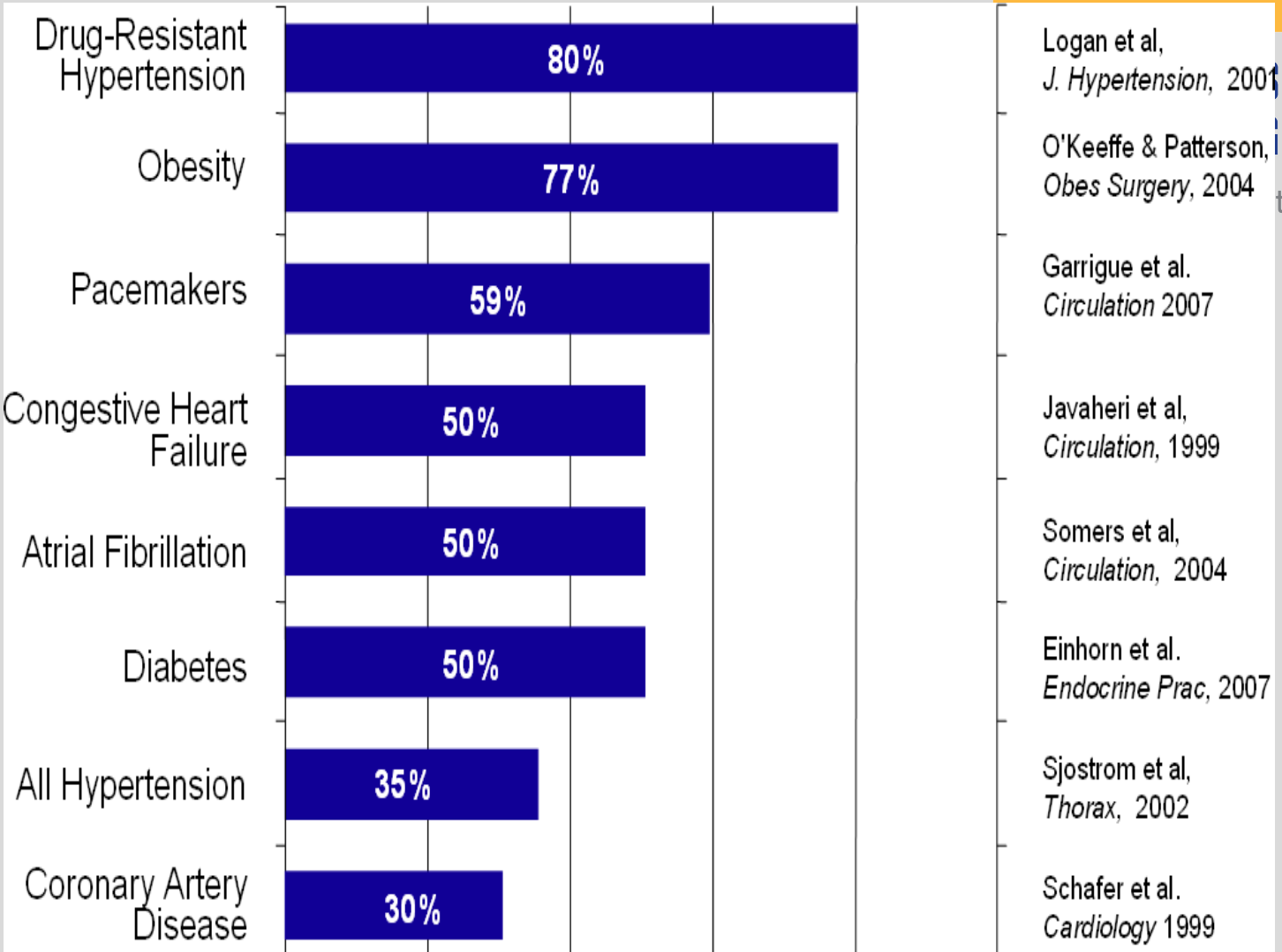
- **Altered sleep architecture:**
  - increased percentage of time spent in stage 1
  - decreased percentage spent in SWS
  - percentage of time spent in rapid eye movement (REM) sleep demonstrates a small (2% to 3%) decrease from middle to old age
  - sleep efficiency lowers with age as well, declining from approximately 86% at about age 45 to about 79% in those older than 70
  - Brief arousals are common in aging and occur at rates of about 15 per hour in the elderly

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Sleep Apnea


Cardiac Problems
Abnormal heart rhythms, heart attack and heart failure
High Blood Pressure
Memory Problems and inability to think correctly
Obstructive Sleep Apnea
Increased insulin resistance
(Even in non-diabetic patients)
Stroke
Increased traffic and workplace accidents


Drug-Resistant Hypertension
80%
Logan et al, J. Hypertension, 2001
Obesity
77%
O'Keeffe & Patterson, Obes Surgery, 2004
Pacemakers
59%
Garrigue et al. Circulation 2007
Congestive Heart Failure
50%
Javaheri et al, Circulation, 1999
Atrial Fibrillation
50%
Somers et al, Circulation, 2004
Diabetes
50%
Einhorn et al. Endocrine Prac, 2007
All Hypertension
35%
Sjostrom et al, Thorax, 2002
Coronary Artery Disease
30%
Schafer et al. Cardiology 1999

INTRODUCTION

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- The term "sleep apnea" encompasses a number of different clinical problems.

- In obstructive sleep apnea (OSA), the most common form of sleep apnea, episodes of apnea occur during sleep as a result of airway obstruction at the level of the pharynx.

EPIDEMIOLOGY

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- A large prevalence study in state employees found that undiagnosed sleep disordered breathing "is prevalent and has a wide range of severity in middle aged women and men"
- In this study, 9.1 percent of men and 4.0 percent of women had apnea/hypopnea indices of 15 or more events per hour.

Young, T, Palta, M, Dempsey, J, et al. The occurrence of sleep-disordered breathing among middle-aged adults. N Engl J Med 1993; 328:1230

EPIDEMIOLOGY

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- It is estimated that, in the United States alone, more than 3 million men and 1.5 million women meet at least one definition of OSA (apnea/hypopnea index of five or more plus a complaint of daytime sleepiness)
- It also appears that clinicians are recognizing OSA in their patients with increasing frequency

PATHOPHYSIOLOGY

- The pharynx, serves a number of functions which seem to conflict, it must be collapsible, for speech and swallowing and must be a conduit for airflow.

- The answer to problem involves a group of muscles that can change the shape of the pharynx when we swallow or talk, but hold it open during inhalation.

PATHOPHYSIOLOGY

- The following physiologic changes are seen with sleep:
  - Reduced tonic input to upper airway muscles
  - Diminished reflexes that protect the pharynx from collapse

PATHOPHYSIOLOGY

- Sleep is associated with pharyngeal narrowing and a substantial increase in inspiratory resistance even in normal individuals.
- An abnormal pharynx can be kept open in wakefulness by an appropriate compensatory increase in dilator muscle activity, but during sleep this compensation fails and the airway collapses.
- Partial collapse results in snoring, hypopneas, and, in some cases, prolonged obstructive hypoventilation. Complete closure results in an apnea.

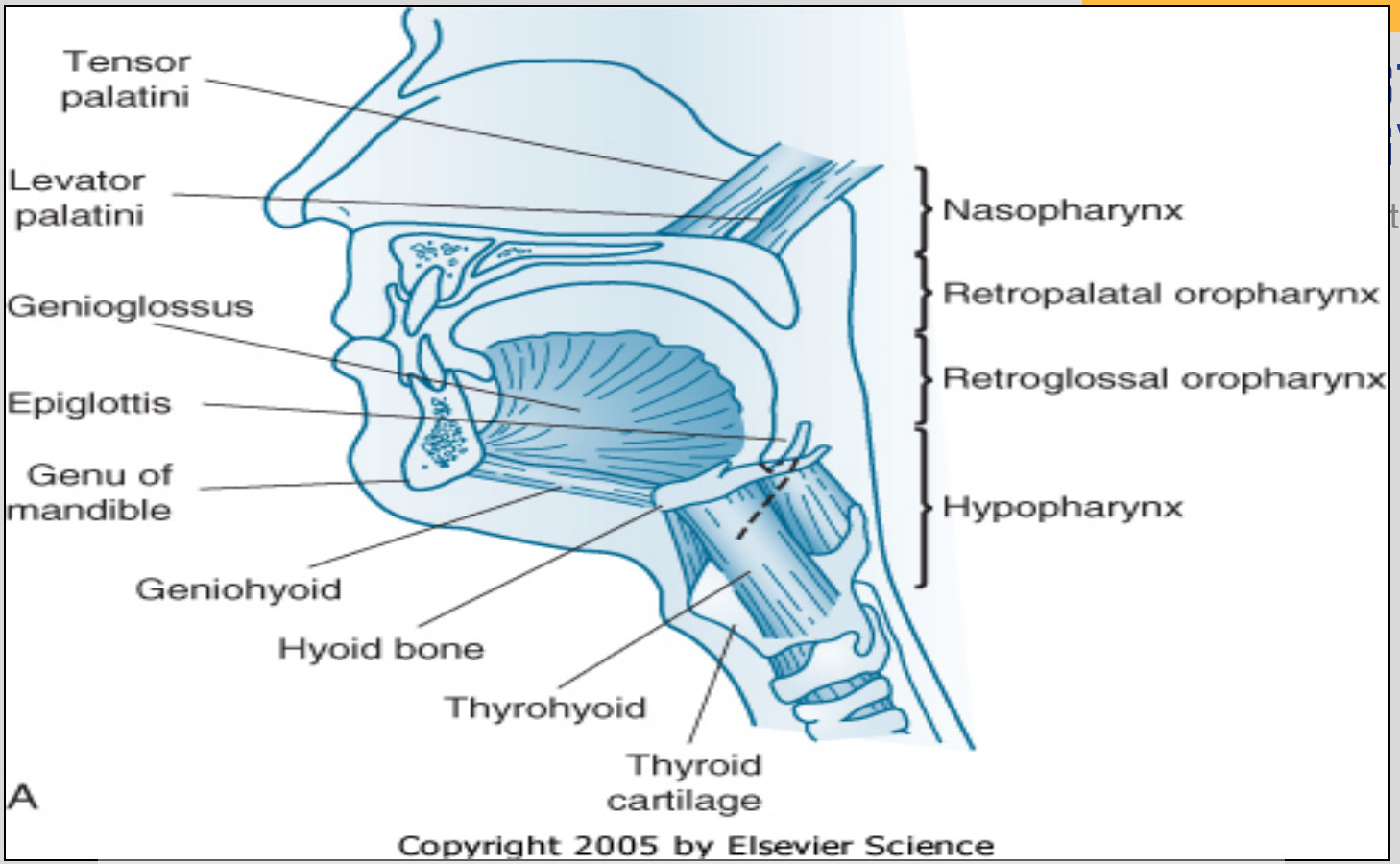


Figure 82-6 A, Schematic diagram of upper airway anatomy. The tensor palatini moves the soft palate ventrally. The genioglossus acts to displace the tongue ventrally. Coactivation of the muscles in the anterior pharyngeal wall such as the geniohyoid and sternohyoid (not shown) act on the hyoid bone to move it ventrally. B, Schematic diagram of upper airway muscles. Among the many upper airway muscles attaching to the floating hyoid bone are the genioglossus, geniohyoid (not shown), hyoglossus, middle pharyngeal constrictor, sternohyoid, and digastric.

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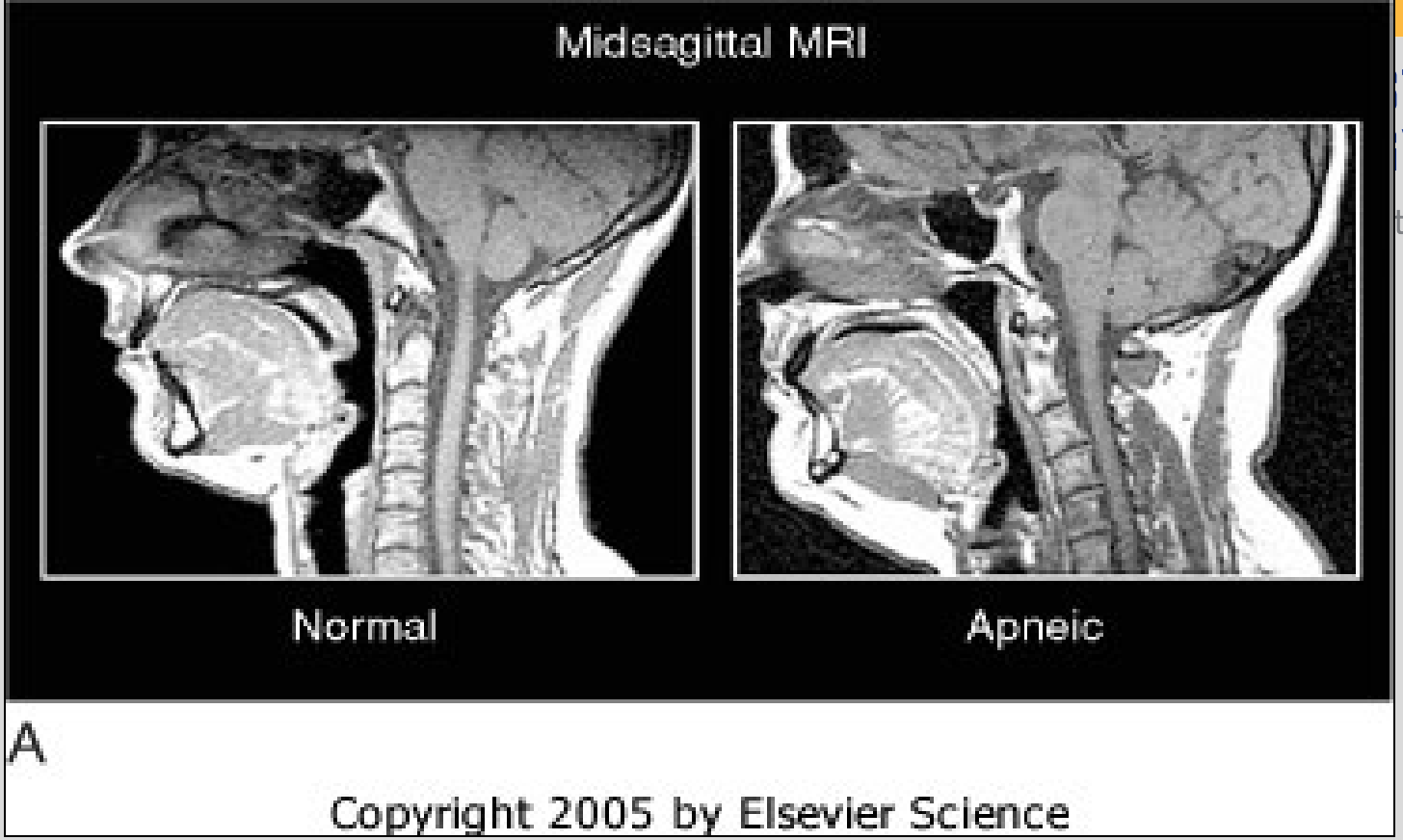


Figure 82-9 A, Midsagittal magnetic resonance (MR) image of a normal subject (left) and a patient with sleep apnea (right). The upper airway is smaller and the soft palate is longer in the patient with sleep apnea. The amount of subcutaneous fat (white area at the back of the neck) is greater in the apneic than in the normal subject. B, Axial MR image in the retropalatal region of a normal subject (left) and a patient with sleep apnea (right). The upper airway is smaller (primarily narrowed in the lateral dimension) in the patient with sleep apnea.

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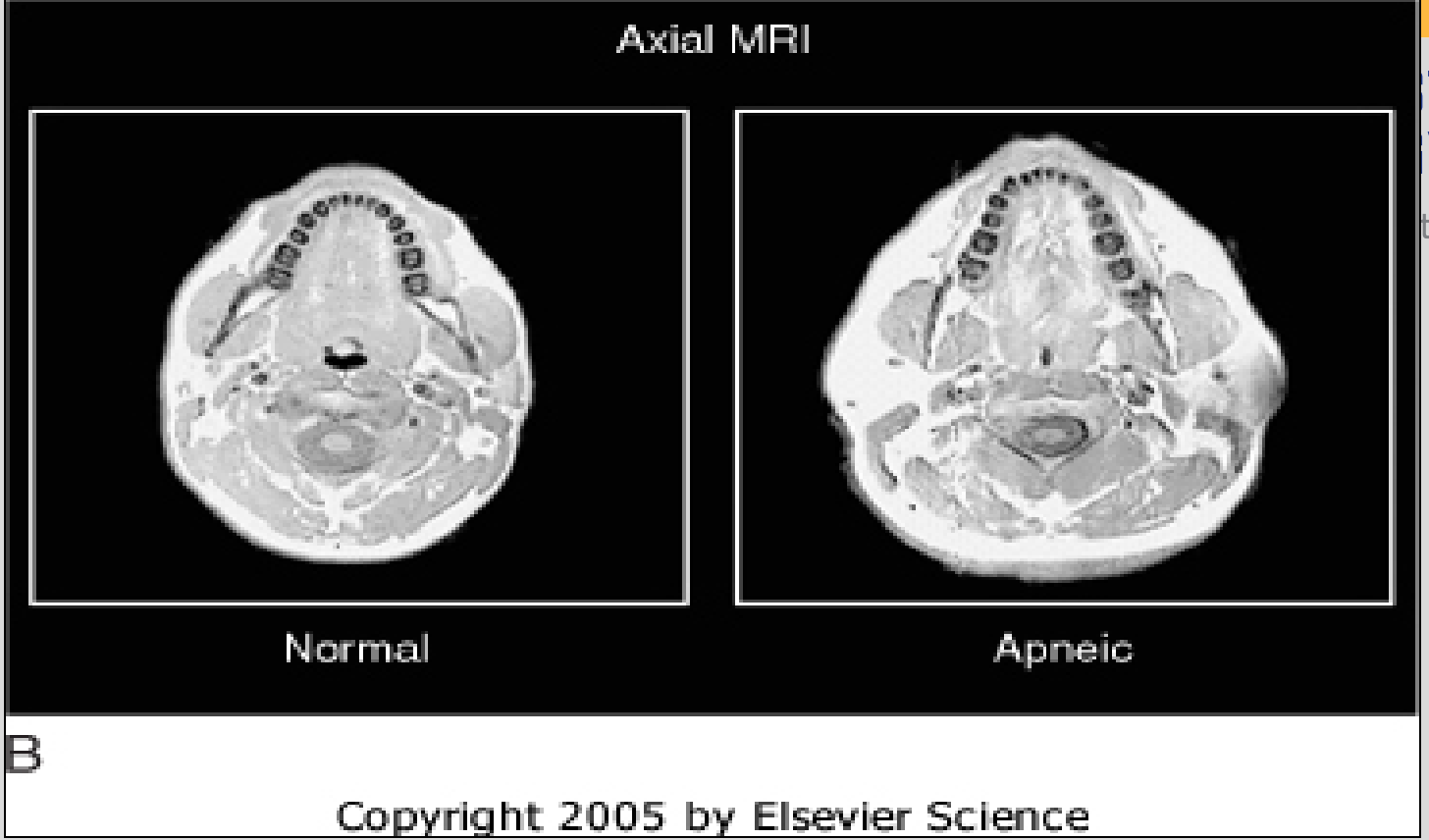


Figure 82-9 A, Midsagittal magnetic resonance (MR) image of a normal subject (left) and a patient with sleep apnea (right). The upper airway is smaller and the soft palate is longer in the patient with sleep apnea. The amount of subcutaneous fat (white area at the back of the neck) is greater in the apneic than in the normal subject. B, Axial MR image in the retropalatal region of a normal subject (left) and a patient with sleep apnea (right). The upper airway is smaller (primarily narrowed in the lateral dimension) in the patient with sleep apnea.

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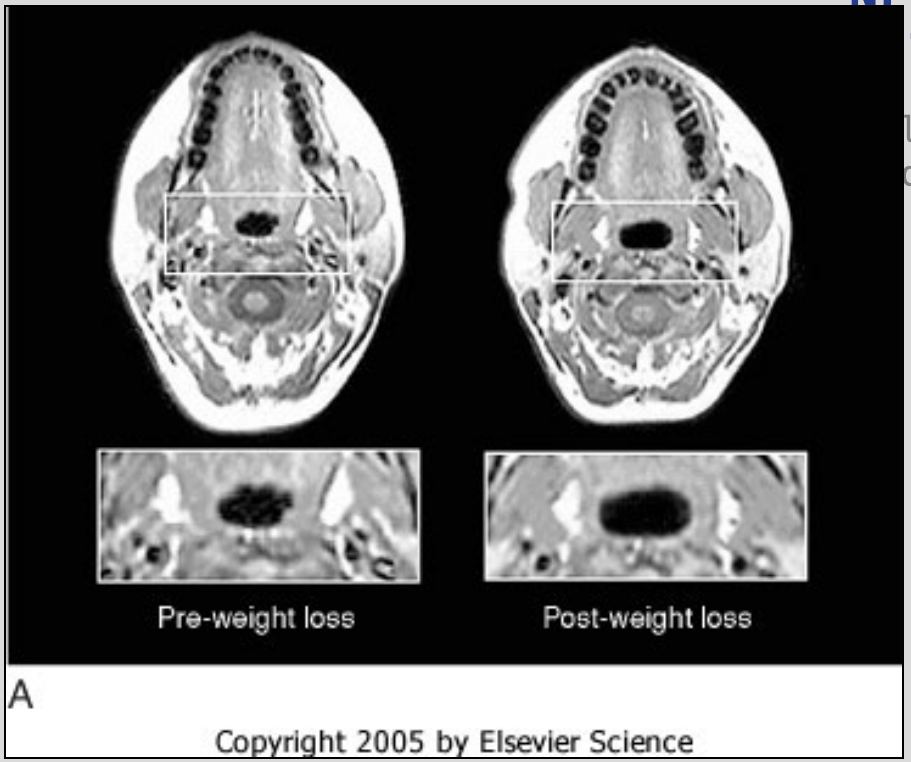
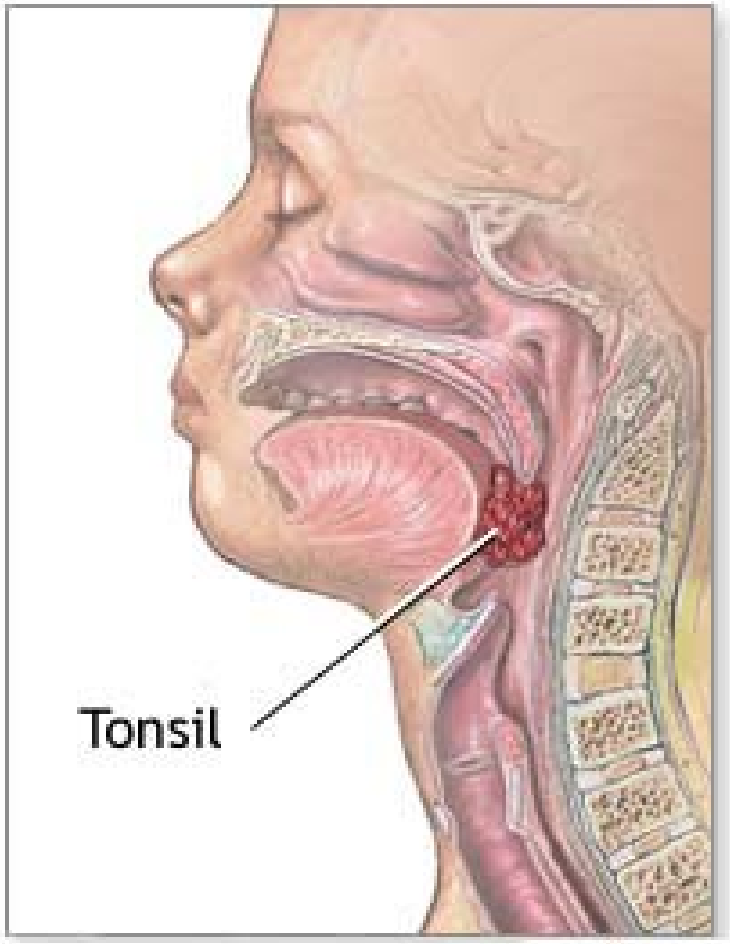


Figure 82-11 A, Axial magnetic resonance images of a normal subject, before and after weight loss in the retropalatal region. Airway area and lateral airway dimensions increase with weight loss. The thickness of lateral pharyngeal walls and the size of the parapharyngeal fat pads decrease with weight loss. B, Volumetric reconstructions of the upper airway soft tissues (soft palate [dark blue], the tongue [light blue], the lateral pharyngeal walls [gray], parapharyngeal fat pads [medium blue]), and craniofacial (mandible [white]) before and after weight loss in a normal subject. The size of the upper airway increases with weight loss. The lateral pharyngeal walls and the parapharyngeal fat pads demonstrated the largest reductions in size with weight loss. Mandibular volume did not change with weight loss.

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Greatly enlarged tonsil causing obstructive sleep apnea



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Obstructive Sleep Apnea Syndrome

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>5 obstructed breaths/hour on PSG

Symptoms:

- Excessive daytime sleepiness
- Loud snoring
- Witnessed apneas
- Choking or gasping during sleep
- Recurrent awakenings from sleep
- Unrefreshed sleep
- Impaired concentration

SLEEP APNEA SYNDROMES

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- **APNEA**
  - CESSATION OF BREATHING > 10 SECONDS

SLEEP APNEA SYNDROMES

- **HYPOPNEA**
  - DECREMENT IN AIR FLOW (50%)
  - ASSOCIATED WITH A 4% DROP IN O2 SAT
  - OR AN EEG AROUSAL

SLEEP APNEA SYNDROMES

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- **APNEA HPOPNEA INDEX (AHI) =**

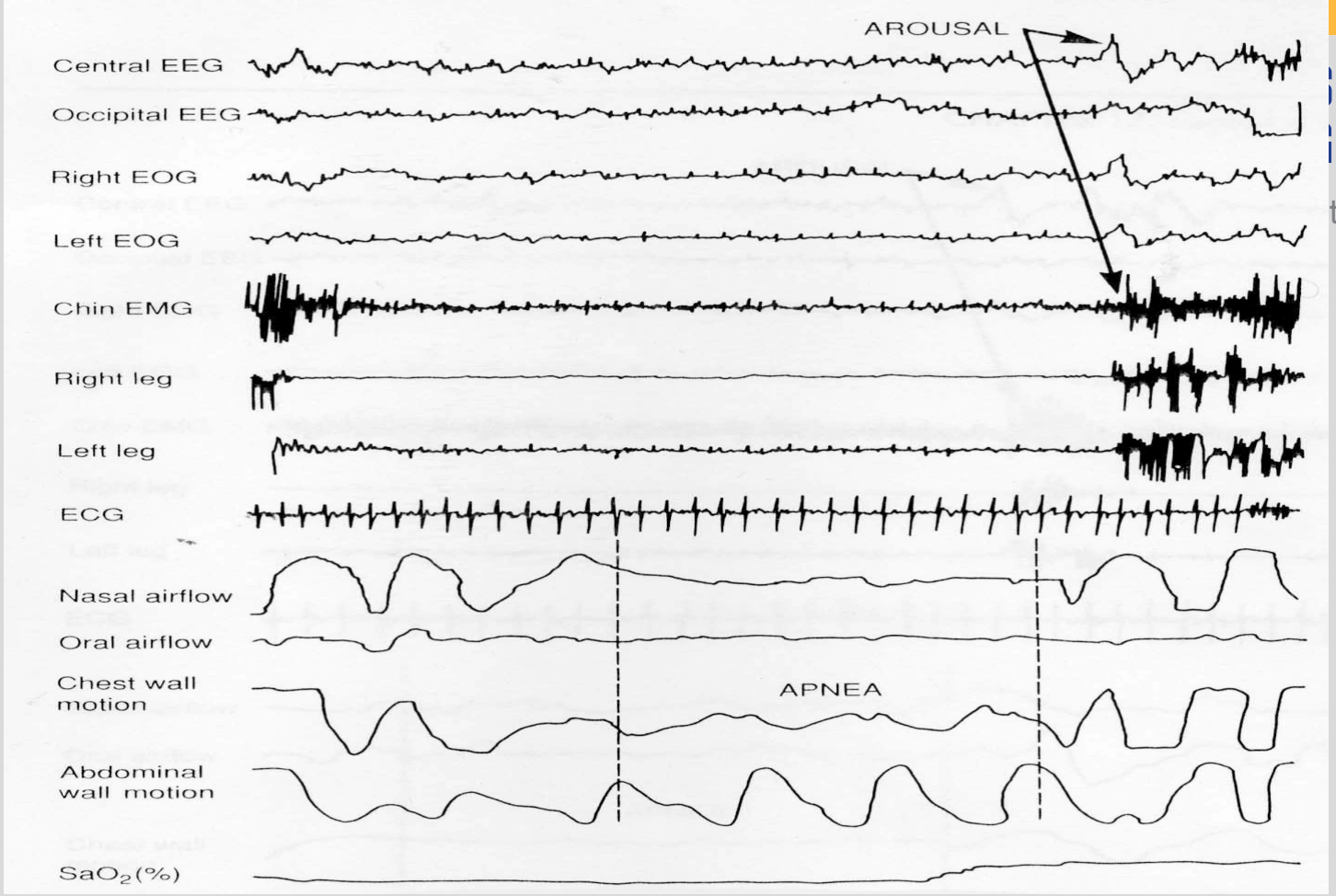
APNEAS + HYPOPNEAS PER HOUR OF SLEEP

- NORMAL APNEA INDEX IS < 5

SLEEP APNEA SYNDROMES

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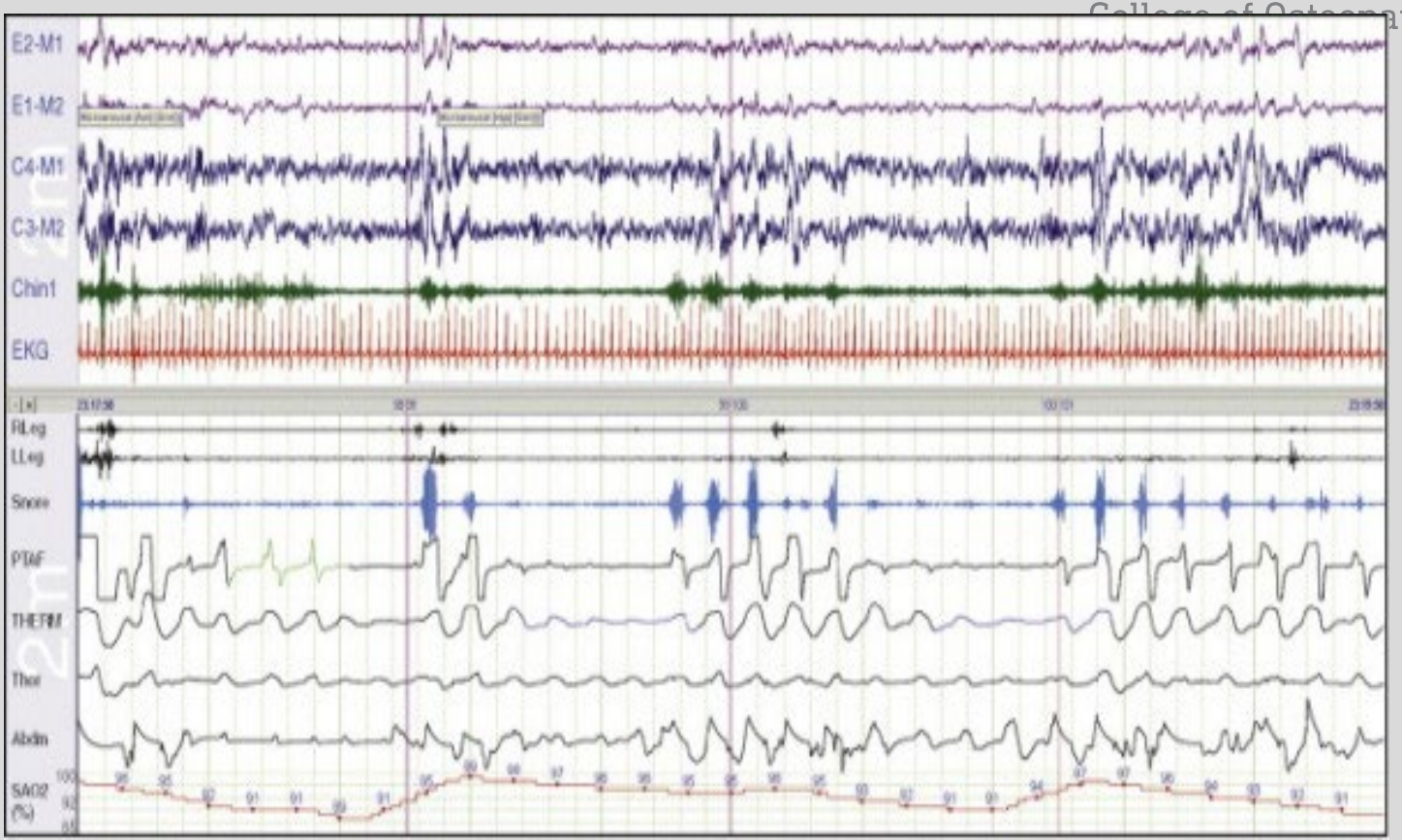
- **THREE TYPES:**
  - OBSTRUCTIVE
  - CENTRAL
  - MIXED



Polysomnogram from a patient with obstructive sleep apnea syndrome

Obstructive sleep apnea

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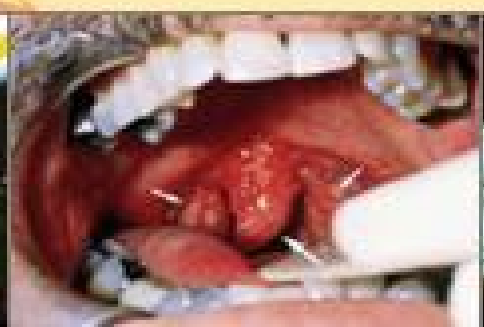
OBSTRUCTIVE SLEEP APNEA (OSA) TREATMENT…

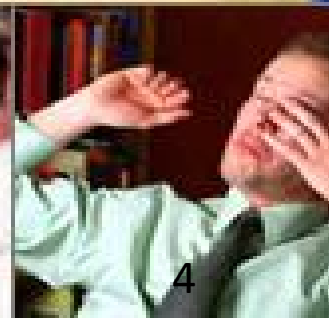
- **Treatment of choice –**
- ***Continuous positive airway pressure (CPAP)***
- Also…
  - Uvulopalatopharygoplasty
  - Dental Device (Oral Appliance)







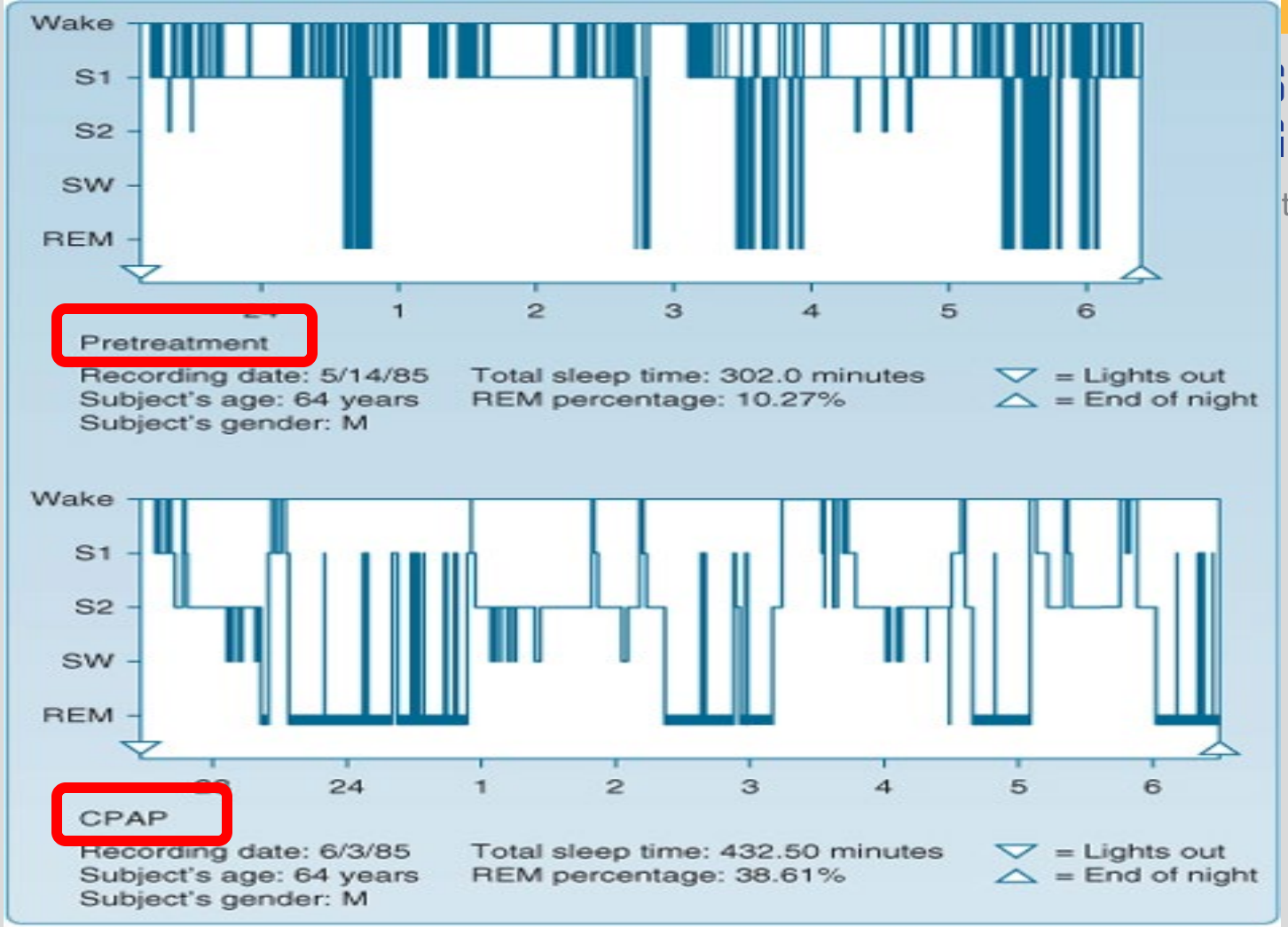






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Figure 90-2 Uvulopalatopharyngoplasty technique. A, Redundant soft palate and tonsillar pillar mucosa are outlined. B, Tonsils, tonsillar pillar mucosa, and posterior soft palate have been excised. The extent of soft palate excision is determined by placing traction on the uvula and noting the position of the mucosal crease. C, Mucosal flaps of the lateral pharyngeal wall and nasal palatal muscle are advanced to the anterior pillar and oral mucosa of the soft palate. The wound is closed with 3-0 Vicryl braided suture. (From Troell RJ, Strom CG: Surgical therapy for snoring. Fed Practitioner 1997;14:29-52.)

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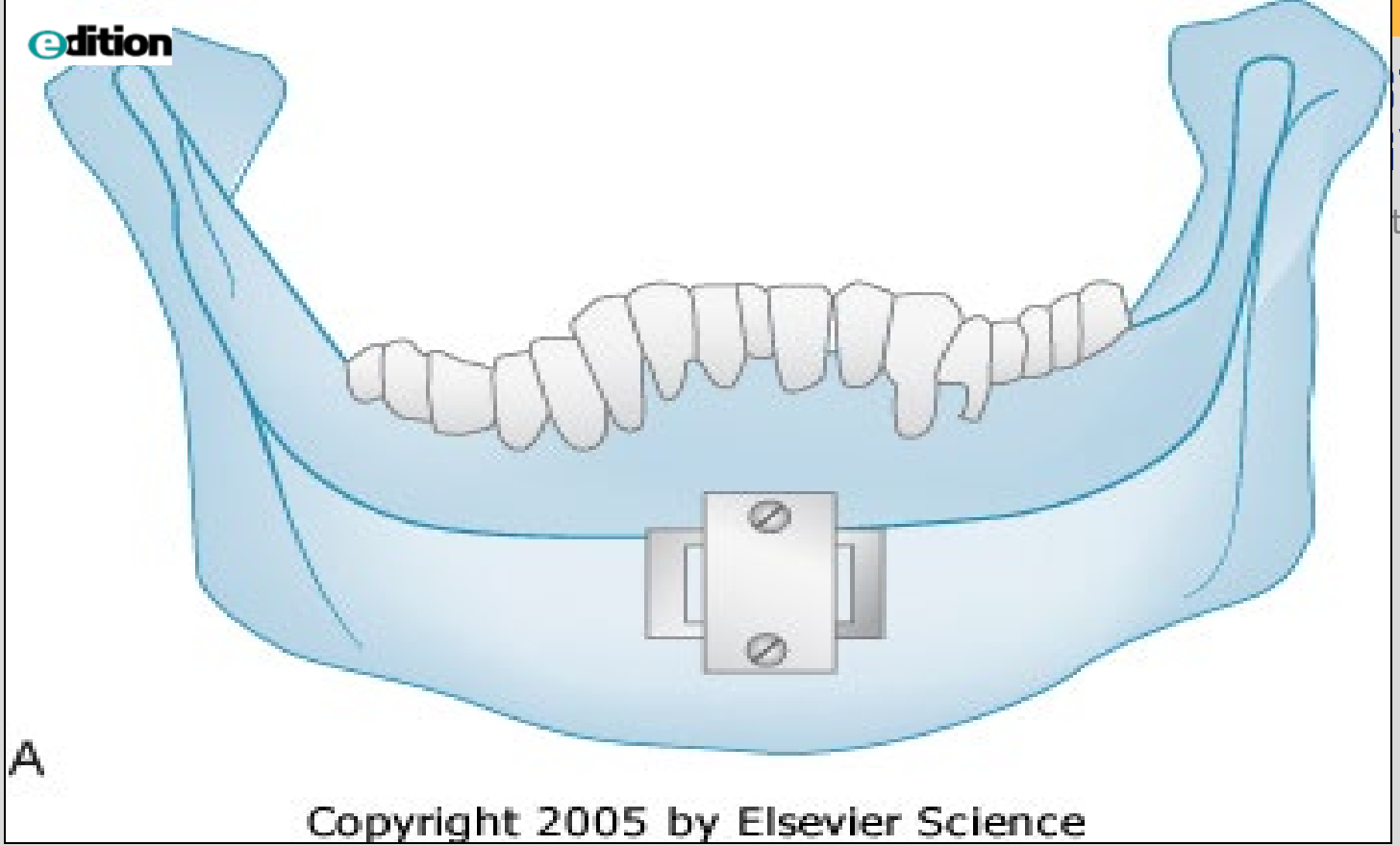


Figure 90-5 Limited mandibular osteotomy with genioglossus advancement procedure (GA). A rectangular window of symphyseal bone consisting of the geniotubercle is advanced anteriorly, rotated to allow bony overlap, and immobilized with a titanium screw. A, Anterior view. B, Lateral view. (From Troell RJ, Powell NB, Riley RW: Hypopharyngeal airway surgery for obstructive sleep apnea syndrome. Semin Respir Crit Care Med 1998;19:175-183.)

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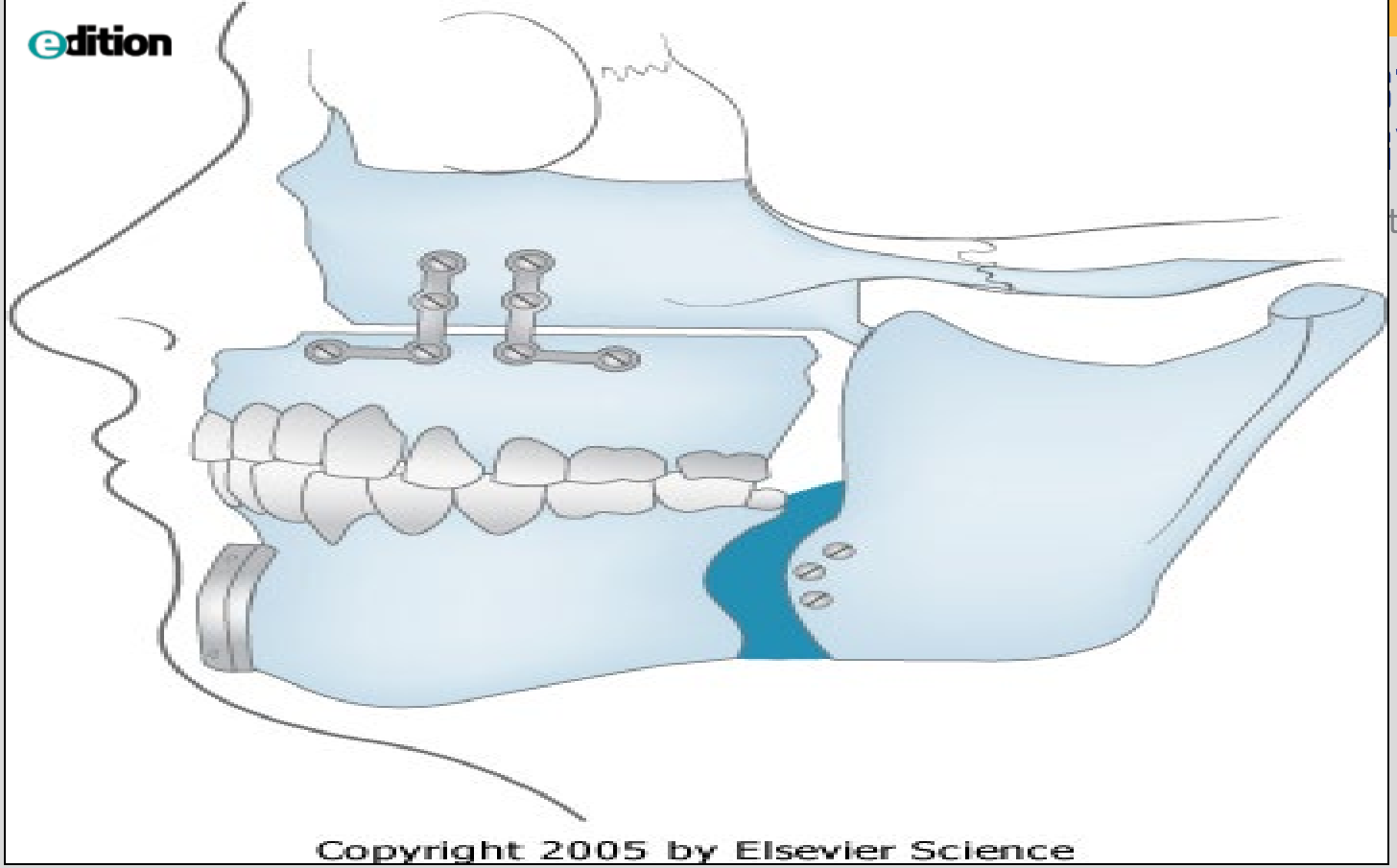
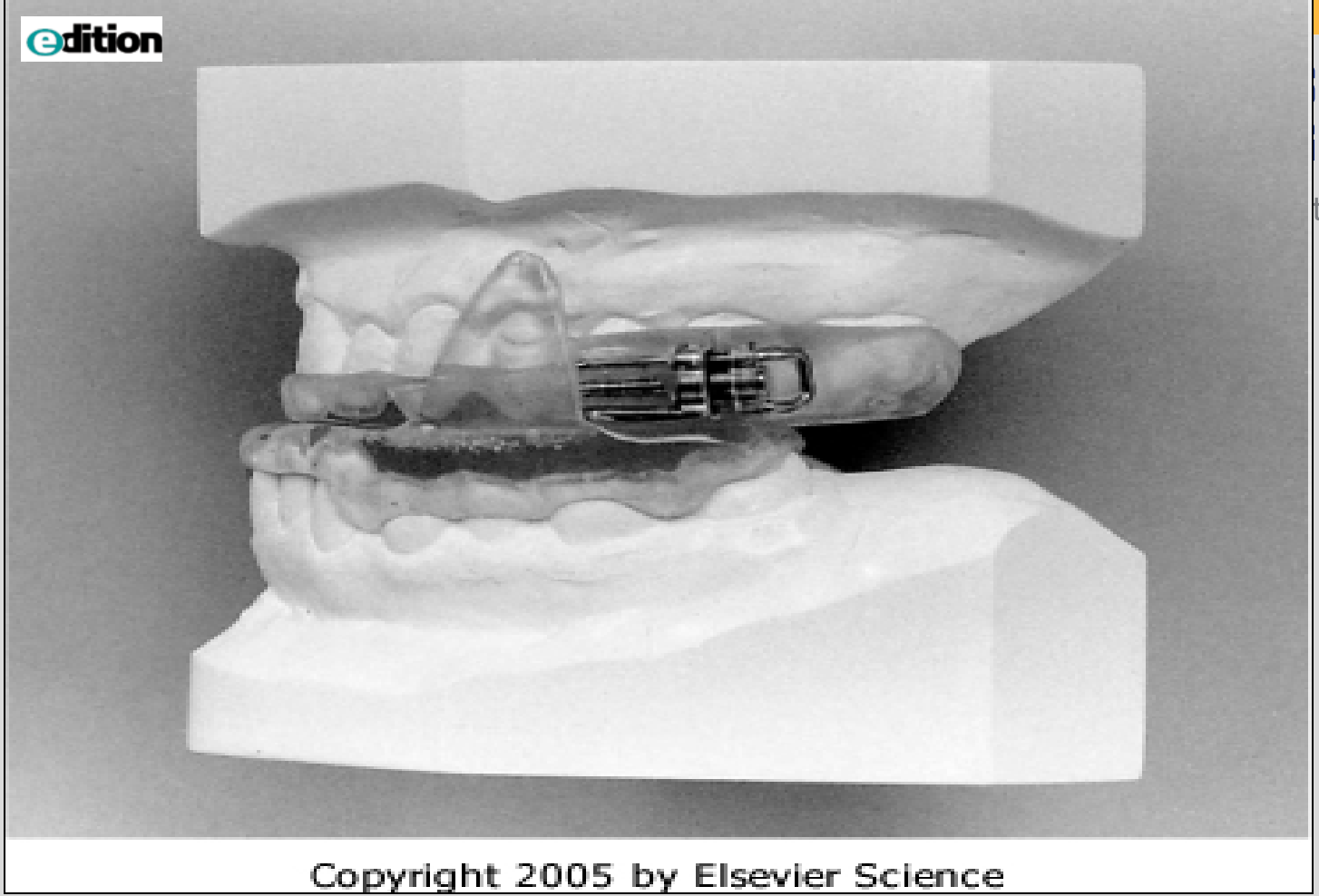


Figure 90-7 Maxillomandibular advancement osteotomy procedure (MMO). Lefort I maxillary osteotomy with rigid plate fixation and a bilateral sagittal split mandibular osteotomy with bicortical screw fixation. The advancement is at least 10 mm. A previous genioglossus advancement is shown. (From Troell RJ, Powell NB, Riley RW: Hypopharyngeal airway surgery for obstructive sleep apnea syndrome. Semin Respir Crit Care Med 1998;19:175-183.)

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Figure 91-6 The mandibular advancement splint. (From Gotsopoulos H, Chen C, Qian J, Cistulli PA: Oral appliance therapy improves symptoms in obstructive sleep apnea: A randomized, controlled trial. Am J Respir Crit Care Med 2002;166:743-748.)

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CENTRAL SLEEP APNEA (CSA)

- REPEATED EPISODES OF APNEA IN ABSENCE OF RESPIRATORY MUSCLE EFFORT
- CENTRAL DRIVE TO THE RESPIRATORY PUMP MUSCLES IS ABOLISHED

CENTRAL SLEEP APNEA (CSA)

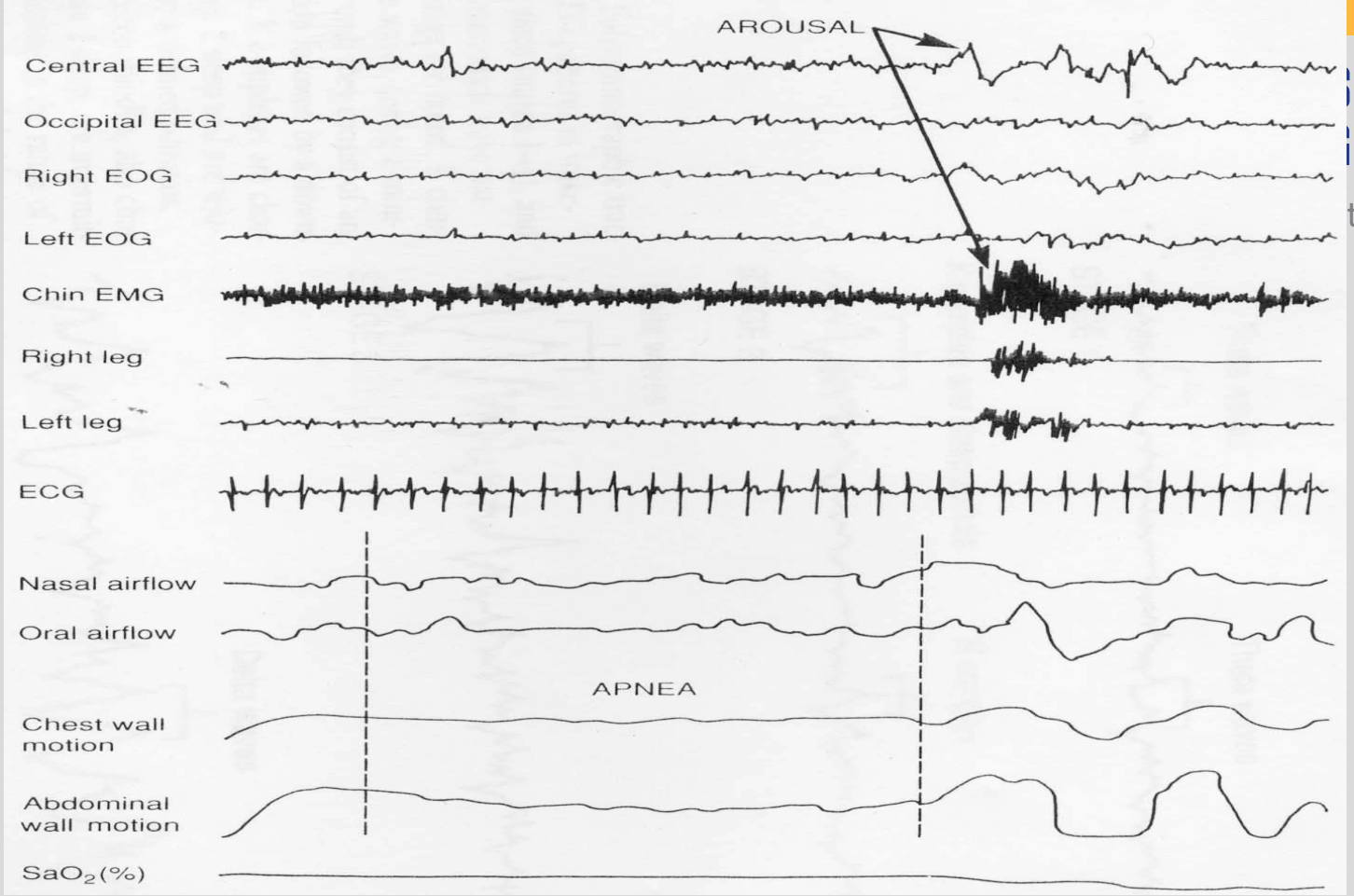
ETIOLOGY:

- IDIOPATHIC
- AUTONOMIC DYSFUNCTION
- NEUROMUSCULAR DISEASE
- BULBAR POLIOMYELITIS
- MYASTHENIA GRAVIS
- ENCEPHALITIS
- BRAIN STEM INFARCTION

CENTRAL SLEEP APNEA (CSA)

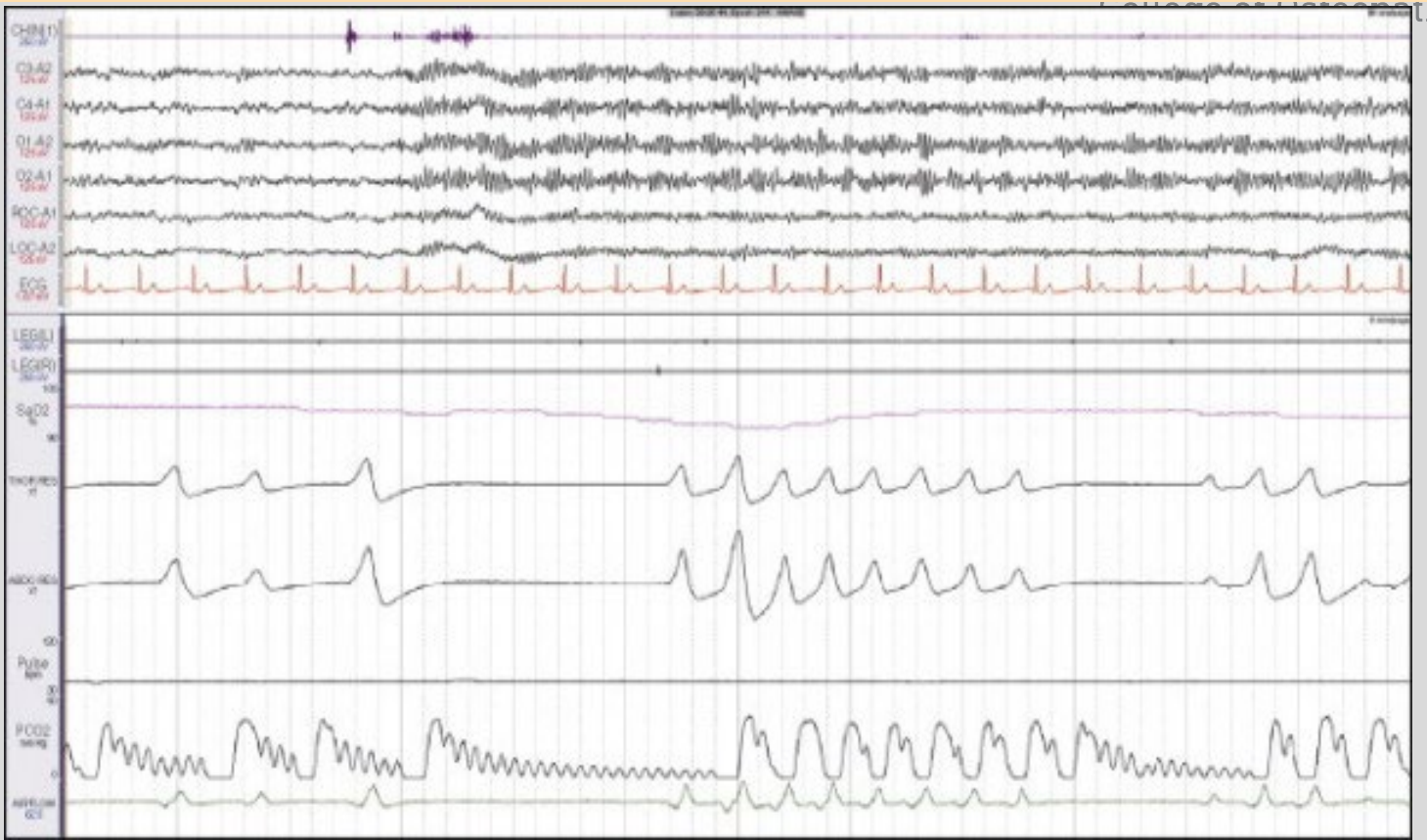
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- PSG - APNEA WITHOUT CHEST WALL OR ABDOMINAL MOVEMENT
- O2 DESATURATION

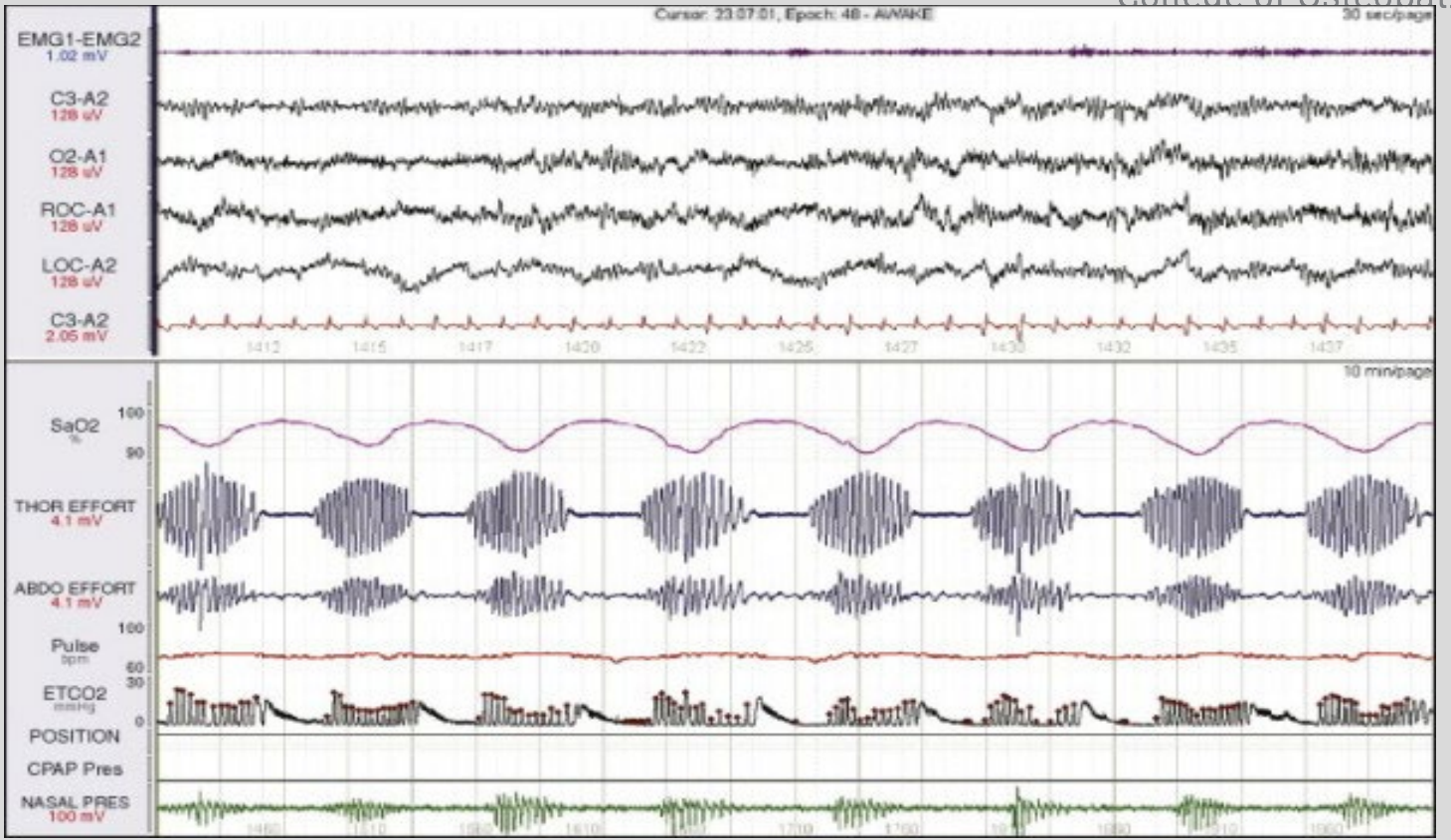


Polysomnogram from a patient with central sleep apnea syndrome.

Central sleep apnea



Cheyne Stokes breathing



CENTRAL SLEEP APNEA (CSA)

- ***Treat the Underlying Problem***
- Treat with Oxygen prn
- CPAP OR BiPAP helpful only 20% of the time.

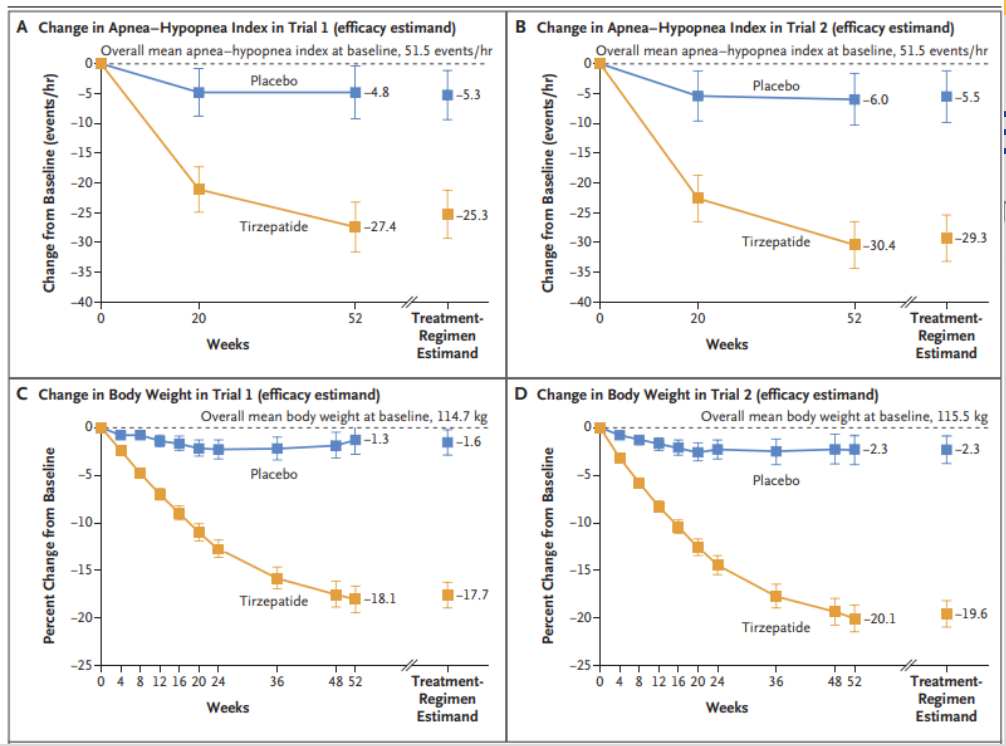
The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

Tirzepatide for the Treatment of Obstructive Sleep Apnea and Obesity

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CONCLUSIONS

Among persons with moderate-to-severe obstructive sleep apnea and obesity, tirzepatide reduced the AHI, body weight, hypoxic burden, hsCRP concentration, and systolic blood pressure and improved sleep-related patient-reported outcomes. (Funded by Eli Lilly; SURMOUNT-OSA ClinicalTrials.gov number, NCT05412004.)

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Restless Leg Syndrome

Restless Legs Syndrome and Periodic Limb Movements during Sleep

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- Neurologic condition characterized by an urge to move, usually associated with paresthesia, that occurs or worsens at rest and is relieved by activity.
- Most patients report difficulty falling asleep or waking up shortly after sleep onset with unpleasant leg sensations
- They also often experience excessive daytime fatigue and somnolence, probably as a consequence of disrupted nocturnal sleep.

Restless Legs Syndrome and Periodic Limb Movements during Sleep

- **Clinical Diagnosis {URGE pneumonic}**
- The urge to move or unpleasant sensations
  - accompanied or caused by uncomfortable and unpleasant sensations in the legs
  - begin or worsen during periods of rest or inactivity such as lying and sitting
  - partially or totally relieved by movement, such as walking or stretching, at least as long as activity continues
  - worse in the evening or night than during the day or occur only in the evening or night

Restless Legs Syndrome and Periodic Limb Movements during Sleep

- **Treatment:**
  - Dopaminergic Agonists: Pramipexole, Ropinirole
  - Dopaminergic Medications: Levadopa
  - Opioids
  - Anticonvulsants: Carbamazepine not useful. Gabapentin showed subjective improvement at doses of 300 to 2400 mg/day
  - Benzodiazepines, including clonazepam, nitrazepam, lorazepam, and temazepam, improve the quality of sleep

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Narcolepsy

Narcolepsy

- Syndrome of unknown origin that is characterized by:
  - Abnormal sleep tendencies
  - Excessive daytime sleepiness
  - Disturbed nocturnal sleep
  - Sleep onset REM periods
  - Cataplexy
  - Sleep paralysis
  - Hypnagogic hallucinations



Narcolepsy

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- The first symptoms often develop near the age of puberty
- The peak age at which reported symptoms occur is 15 to 25 years
- A second, smaller peak of onset has been noted between 35 and 45 years and near menopause in women
- There is often a marked delay of over 10 years before diagnosis is made, especially if cataplexy is initially absent

Narcolepsy

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- **Diagnosis:**
  - The nocturnal polysomnogram may indicate the underlying cause of sleepiness
  - MSLT indicates the severity of the problem
  - Once the nocturnal sleep recording has eliminated specific diseases, the MSLT confirms the diagnosis of narcolepsy with the presence of two or more sleep-onset REM periods.

Narcolepsy

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- **Diagnosis:**
  - The ***multiple sleep latency test (MSLT)*** was designed to measure physiologic sleep tendencies in the absence of alerting factors
  - It consists of five scheduled naps, usually at 9 am, 11am, and 1, 3, and 5 pm
  - The subject is polygraphically monitored in a comfortable, soundproof, dark bedroom, while wearing street clothes

Narcolepsy

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- **Diagnosis:**

- Genetic testing: human leukocyte antigen DQB1*0602

- In most patients with narcolepsy, there is an absence or very low level of **hypocretin** in the cerebrospinal fluid

Treatment for Narcolepsy –

for reference

General Measures

Avoid shifts in sleep schedule
Avoid heavy meals and alcohol intake
Regular timing of nocturnal sleep: 10:30 PM to 7 AM
Naps: Strategically timed naps, if possible (e.g., 15 min at lunchtime, 15 min at 5:30 PM)

Medication for Sleepiness

Modafinil
Methylphenidate
GHB (sodium oxybate)
Dextroamphetamine sulfate

Medication for Cataplexy

Fluoxetine 20-60 mg
Venlafaxine 150-300 mg
GHB (see above)
Clomipramine 75-125 mg
Viloxazine 150-200 mg
Imipramine 75-125 mg

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Insomnia

CRITERIA FOR DEFINING INSOMNIA

- **Difficulty falling asleep, *or***
- **Difficulty staying asleep, *or***
- **Early-morning awakening, *or***
- **Nonrefreshing or nonrestorative sleep.**

With daytime consequences or distress lasting at least 1 month despite opportunity and circumstances to sleep.

The daytime consequences of insomnia include impaired cognitive function, fatigue or tiredness, depressed mood or irritability, and, in adults, impaired performance at work, and, in children, poor performance in school.

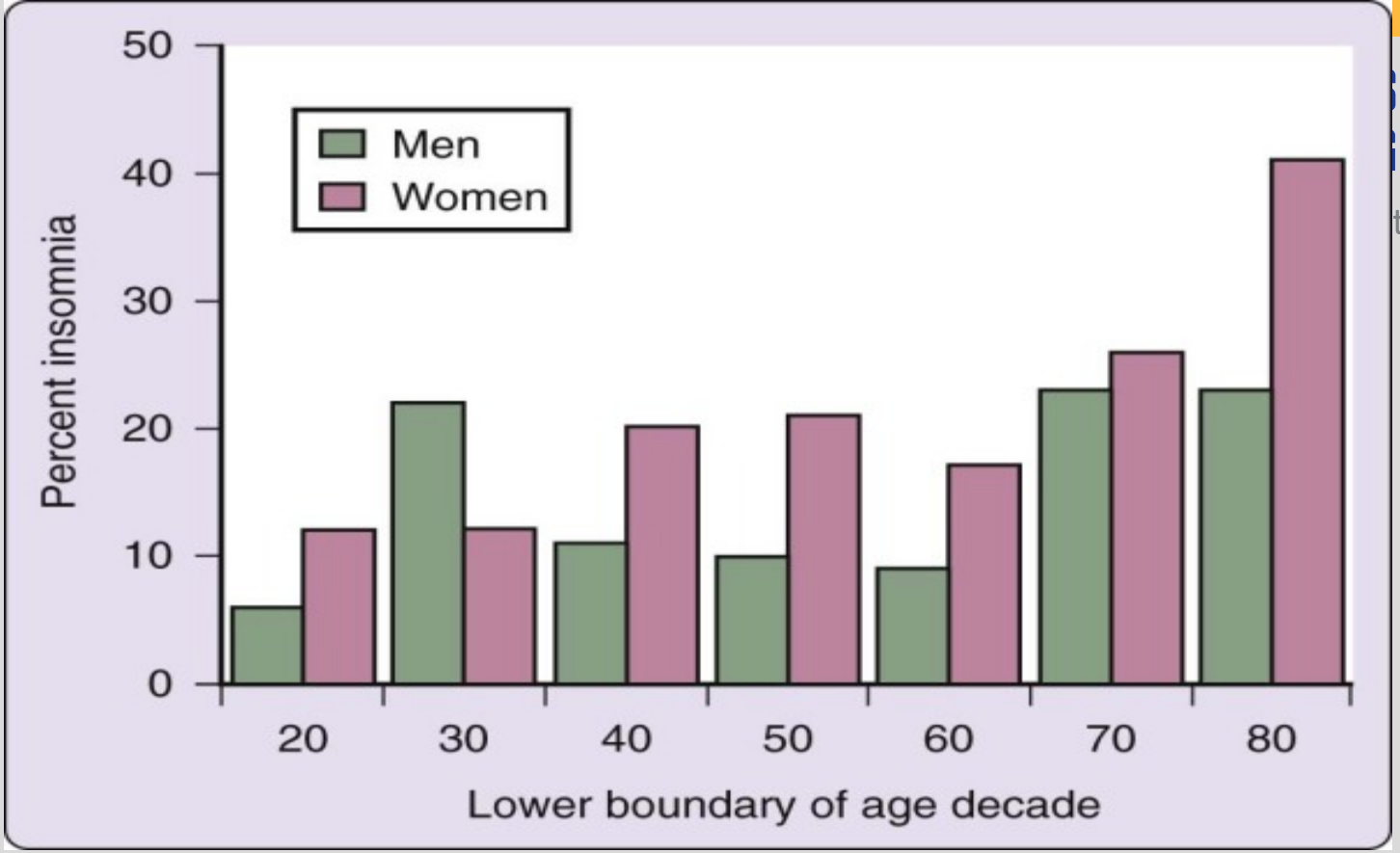
Epidemiology

About 30% to 40% of adults have some degree of disturbed sleep during the year.

About 10% to 13% meet diagnostic criteria for insomnia.

The most important risk factors for insomnia are:

- Gender
- Age
- Economic and Socioeconomic Status



Insomnia prevalence by sex and age. Each value on the abscissa represents the beginning of an age decade as in 20 (to 29) and 30 (to 39). *(Data from Lichstein KL, Durrence HH, Riedel BW, et al. Epidemiology of sleep: age, gender, and ethnicity. Mahwah, NJ: Erlbaum; 2004.)*

Risk Factors

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Static Risk Factors:

- Gender
- Age
- Ethnicity
- Genetics

Modifiable Risk Factors:

- Hyperarousal
- Stress and life events
- Transient insomnia
- Medical comorbidities
- Psychological comorbidities

COMORBID INSOMNIA

Comorbid insomnia refers to patients who have a medical or psychiatric condition and who also have insomnia symptoms

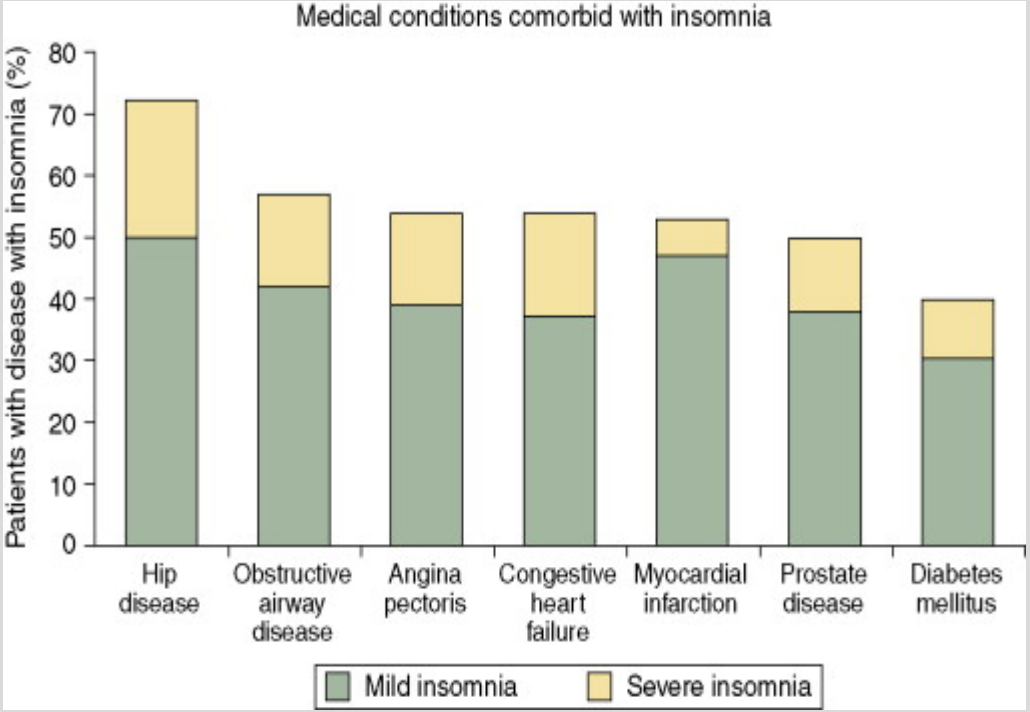


Figure 9-9 In a study of 3445 patients with chronic medical conditions a substantial proportion of patients with common comorbidities had mild or severe insomnia even in the absence of depression. The mean age of the studied population of patients was 54 years. ***(Data from Katz DA, McHorney CA: Clinical correlates of insomnia in patients with chronic illness. Arch Intern Med 158:1099–1107, 1998.)***

COMORBID INSOMNIA

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MEDICAL CONDITIONS COMORBID WITH INSOMNIA

- Cardiovascular disease (CHF and stroke)
- Pulmonary disease (asthma, COPD)
- Gastrointestinal (PUD/GERD)
- Endocrine (diabetes mellitus)
- Renal (renal failure)/Urologic (nocturia)
- Neurologic (Alzheimer's, Parkinson's)
- Pain from any source (arthritis, fibromyalgia)
- Perimenopause and menopause
- Cancer

PSYCHIATRIC CONDITIONS COMORBID WITH INSOMNIA

- Anxiety disorder
- Depression
- Schizophrenia
- Alcohol and drug abuse

Treatment of Choice: Cognitive-Behavioral Therapy

- Sleep hygiene
- Stimulus control
- Sleep Restriction
- Relaxation
- Cognitive

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Thank You…

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